

REVIEW

Curcumin: A Potential Detoxifier Against Chemical and Natural Toxicants

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ABSTRACT

The human body gets exposed to a variety of toxins intentionally or unintentionally on a regular basis from sources such as air, water, food, and soil. Certain toxins can be synthetic, while some are biological. The toxins affect the various parts of the body by activating numerous pro-inflammatory markers, like oxidative stresses, that tend to disturb the normal function of the organs ultimately. Nowadays, people use different types of herbal treatments, viz., herbal drinks that contain different spices for detoxification of their bodies. One such example is turmeric, the most commonly available spice in the kitchen and used across all kinds of households. Turmeric contains curcumin, which is a natural polyphenol. Curcumin is a medicinal compound with different biological activities, such as antioxidant, antineoplastic, anti-inflammatory, and antibacterial. Hence, this review gives a comprehensive insight into the promising potential of curcumin in the detoxification of heavy metals, carbon tetrachloride, drugs, alcohol, acrylamide, mycotoxins, nicotine, and plastics. The review encompasses diverse animal-based studies portraying curcumin's role in nullifying the different toxic effects in various organs of the body (especially the liver, kidney, testicles, and brain) by enhancing defensive signaling pathways, improving antioxidant enzyme levels, inhibiting pro-inflammatory markers activities and so on. Furthermore, this review also argues over curcumin's safety assessment for its utilization as a detoxifying agent.

1 | Introduction

Modern chemical technologies have been applied in nearly all human activity spheres along with a wide array of diverse chemicals (Lysiuk et al. 2020). The high toxic potential of these chemicals, their capability to pollute the environment and their

penetration into the human body through different means can result in a higher risk of acute as well as chronic mass poisoning (Lysiuk et al. 2020). Contamination of the environment now presents a major global concern, posing adverse effects on food security and ecosystem health. Air pollution alone has caused millions of premature foetal deaths annually and globally,

predominantly associated with chronic obstructive pulmonary disease (COPD), lung cancer, asthma, stroke, respiratory infections, and heart failure, as reported by the World Health Organization (WHO) (Demir and Kacew 2023). Data suggests that 99% of the human population breathes contaminated air at levels above the recommended limits. Recently, it has been summarized that overdose of drugs associated with higher alcohol as well as alcoholic beverages consumption, a fat-rich diet, and recurrent viral infections act as major contributors to the development of various liver disorders in human beings (Bridi, von Poser, and de Carvalho Meirelles 2023). Plastics, including microplastics, are generally regarded as harmful to organisms owing to their physical characteristics (Weis and Alava 2023). The plastic particles are taken up by the organisms readily via respiration and ingestion and are reported throughout food webs. Extensive studies have reported the effects of plastics on an array of animals and plants and health effects at genetic, biochemical, and organismal levels (Weis and Alava 2023). Food contaminants constitute all types of unwanted dietary substances produced from the cultivation conditions applied, environmental exposure or production processes (Lootens et al. 2022). Some contaminants pose a threat to human health by producing toxic secondary metabolites such as mycotoxins derived from fungi (Lootens et al. 2022). These toxins have also been detected in various food crops, including maize, sorghum, peanuts, and wheat (Lootens et al. 2022). They are produced as a defence mechanism under stressful conditions in fungi and are toxic to animals and humans, causing illness and occasional death as well (Lootens et al. 2022).

From an integrative medicine viewpoint, there are diverse approaches to removing toxins from the body, the most accepted and common approach being the utilization of herbal preparations (Bjørklund et al. 2024). Nowadays, the global paradigm for health promotion has drifted swiftly 'from pharmaceuticals towards nutraceuticals' because of the high cost and several side effects arising due to the consumption of pharmaceuticals, especially synthetic ones (Bjørklund et al. 2024). The utilization of detoxifying agents derived from herbal raw materials can be justified if given in correct dosing and for sufficient duration, not displaying inferiority to synthetic molecules in terms of effectiveness and having a significantly improved safety profile. Curcumin is a well-known and naturally occurring polyphenolic compound isolated from *Curcuma longa* or turmeric rhizome (Ashrafizadeh et al. 2020). Curcumin is chemically known as diferuloylmethane and has a molecular formula of $C_{21}H_{20}O_6$. It is the most significant natural compound that has been isolated from plants and investigated recently (Dorcas et al. 2021). This medicinal compound displays different biological activities, such as antioxidant, antineoplastic, anti-inflammatory, and antibacterial. It also shows therapeutic effects on renal disorders, diabetes mellitus, and neurodegenerative disorders. Curcumin has been globally recognized and used in various forms for its health benefits. In India, it is an important ingredient in curries; in Japan, it is served as turmeric tea; in Thailand, cosmetics containing curcumin are popular; in China, it serves as a colouring agent; in Korea, it is included in various beverages; in Malaysia, it is applied as an antiseptic; in Pakistan, its anti-inflammatory properties are valued; and in the US, it is used in mustard sauce, butter, chips, and cheese as a coloring agent and preservative, in addition to being available in powder form and capsules (Hewlings

and Kalman 2017). Commercially, curcumin is available in various forms, such as capsules, ointments, tablets, energy drinks, cosmetics, and soaps (Gupta, Patchva, and Aggarwal 2013). The United States Food and Drug Administration (USFDA) recognizes curcuminoids as "Generally Recognized As Safe" (GRAS) (Gupta, Patchva, and Aggarwal 2013).

Curcumin is well-tolerated and safe at high concentrations with no evidence of toxicity. Web of Science (WOS) data between the year 2011 and 2023 includes "curcumin and health" as the search string, with the publication of nearly 2216 peer-reviewed journal articles associated with curcumin and health so far. The complete record of articles, including their citations, has also been exported in plain text format. VOSviewer (version 1.6.19), a software tool, was further utilized to map down and inspect the keywords for co-occurrence analysis. Furthermore, Figure 1 displays keyword co-occurrence analysis describing oxidative stress, antioxidant, inflammation, bioavailability, pharmacokinetics, stability in vivo, and transcription factor, nuclear factor kappa-light-chain-enhancer of activated B cells (NF- κ B) acting as key areas to connect curcumin. This review brings together the known information on the worldwide burden posed by natural and chemical-induced toxicity, especially stemming from substances such as heavy metals, plastics, acrylamide, carbon tetrachloride (CCl_4), alcohol, nicotine, mycotoxins, and drugs. Delving into exhaustive details, this review illuminates the antitoxic property of curcumin as evidenced by the in vivo studies. Additionally, the review illustrates the significance of curcumin safety assessments for the development of the medicinal compound into a safe detoxifier agent.

2 | Global Burden of Chemical and Naturally-Induced Toxicity

Figure 2 illustrates a schematic representation of diverse toxicants impacting the vital body organs utilizing animal models.

2.1 | Heavy Metals

2.1.1 | Arsenic (As)

Arsenic is a widely known toxic metalloid that exists naturally inside the earth's crust in the form of organic and inorganic compounds (Shayan et al. 2023). Even though exposure to arsenic can occur through medicinal, environmental, nutritional, and industrial means, aquifers and water sources are assumed to be the utmost underlying reason for arsenic poisoning (Mochizuki 2019). Usually, arsenic exposure appears as inorganic pentavalent (As^{5+}) and trivalent (As^{3+}) forms, which are considered more toxic and more potent than their organic forms (Medda et al. 2020). Inside the body, the inorganic arsenic is first converted into its organic metabolites through oxidative methylation and glutathione conjugation and gets excreted through bile and urine (Hayakawa et al. 2005; Pinyayev et al. 2011). Arsenic can invite severe outcomes for different organs (Vahidnia, van der Voet, and de Wolff 2007; Lengfelder, Hofmann, and Nowak 2012; Sharma and Kumar 2019). Pulmonary and cardiovascular toxicity, reproductive damage, carcinogenicity, and gastrointestinal and renal injuries occur as a consequence

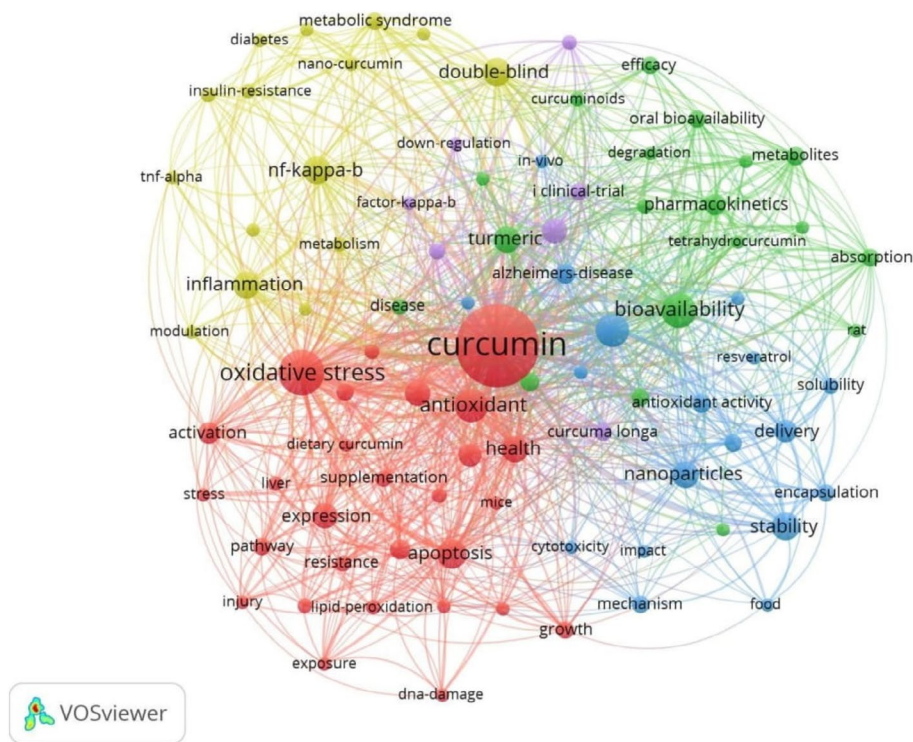


FIGURE 1 | Visual drawing exploring keyword co-occurrence analysis of curcumin.

of arsenic exposure, along with neurotoxicity, which remains the central theme of this study (Vahidnia, van der Voet, and de Wolff 2007; Lengfelder, Hofmann, and Nowak 2012; Sharma and Kumar 2019) (Table 1).

In Asia, contamination is a serious concern, with 31 countries being reported for severe impact. The global affected population proportion is 94% (major contribution by Asia), and the exposure of millions of inhabitants to high arsenic levels in drinking water occurs mainly due to natural contamination. This has led to critical health crises, with the prevalence of arsenic-induced skin lesion cases and an increased risk of a variety of cancers (Bhat et al. 2024). In Argentina, around 4 million people have been exposed to excessive arsenic amounts in the drinking water. Higher amounts ($>1000\mu\text{g/L}$) of arsenic have been recognized in the Cuyo regions of the Andes and Chaco-Pampean Plains, located in the northwestern and central regions of the country, respectively (McClintock et al. 2012). However, contamination is not as widespread in the United States of America (USA) as in other countries. However, certain areas, including parts of Southwest America, have been reported to have elevated levels of arsenic in groundwater, attributed mainly to geological factors. In Western USA, states like California and Arizona have been historically associated with mining activities, and hence, arsenic pollution affects both local communities and water sources, highlighting the significance of the ongoing remediation efforts and stringent regulations in the USA, wherein over 2 million individuals utilize drinking water drawn from private wells containing arsenic concentrations exceeding the permissible regulatory limit ($10\mu\text{g/L}$) (Bhat et al. 2024). Similarly, the northeastern region of Hungary also faces arsenic contamination problems in groundwater sources (Bhat et al. 2024). Arsenic contamination is also a matter of concern in Australia and is primarily associated with industrial and mining activities.

The contaminated areas include regions of Queensland, South Australia, and Western Australia, where mining operations tend to release arsenic into the environment, thus affecting both groundwater and surface sources (Bhat et al. 2024). In regions of Africa, arsenic contamination also poses a considerable health risk, with 20 countries being affected (Irunde et al. 2022).

2.1.2 | Lead (Pb)

Lead has been enlisted among the 10 chemicals that are raising major public health challenges by the WHO. Its exposure ranks fourth amongst key environmental health risk issues, after air pollution due to ambient particulate matter, air pollution arising from solid fuels used in households, and unsafe drinking water in households, handwashing, and sanitation, based upon the 2019 Global Burden of Diseases, Injuries, and Risk Factors Study (GBD) (Larsen and Sánchez-Triana 2023). Every year, approximately 1 million individuals die from lead poisoning, and many more than this number, including children, get exposed to lower lead levels, thus resulting in lifelong health problems, such as anaemia, immunotoxicity, reproductive toxicity, and hypertension. The behavioural and neurological effects produced by lead could not be reversed (WHO 2022). According to UNICEF estimates, every 1 in 3 children, with the number rising to 800 million worldwide, have been detected with blood lead levels $\geq 5\mu\text{g/dL}$. Hence, immediate global attention and action are requisite to address this issue (WHO 2022). Data collected from a GBD study between 1990 and 2019 suggests an increase in the disability-adjusted life years (DALYs) and mortality rate resulting from exposure to lead by 35.26% and 70.19%, respectively, has been noticed; however, a decrease in the age-standardized DALYs rate (ASDR) and age-standardized mortality rate (ASMR) by 29.23% and 20.66%, respectively have been

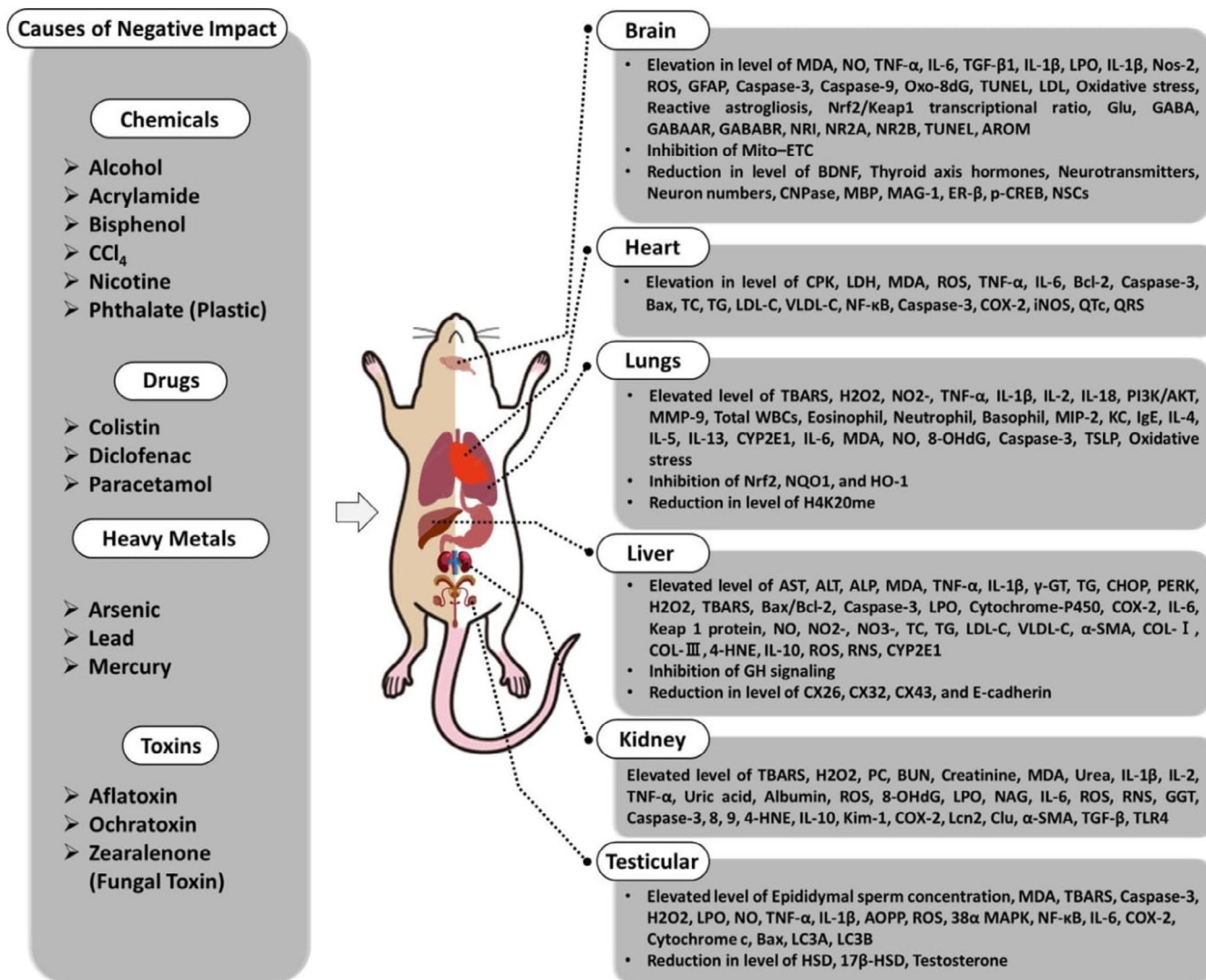


FIGURE 2 | An overall presentation of negative consensuses of chemical and natural toxicants on different vital organs.

observed. Ischemic heart disease (IHD), hypertensive heart disease (HHD), and stroke displayed the highest increment in deaths; stroke, diabetes and kidney disease (DKD), and IHD showed the fastest-growing DALYs (Table 1) (Xu et al. 2023).

2.1.3 | Mercury (Hg)

Mercury ranks third on the list of most toxic elements for human health, according to the Agency for Toxic Substances and Disease Registry (ATSDR) under the USA government (Budnik and Casteleyn 2019). Pure Earth approximations suggest that a global health risk is associated with mercury exposure among 19 million people, with nearly 1.5 million DALYs of disease burden (Budnik and Casteleyn 2019). Toxic Sites Identification Program in 2015 identified more than 450 sites across the world wherein mercury exposure tends to threaten the population's health (Budnik and Casteleyn 2019). With respect to mercury emissions, a substantially large population is affected. Several studies have attempted to assess the financial burden methyl mercury poses on prenatal exposure, the associated intelligence quotient (IQ) loss, and, consequently, the lowering of production and earning potential. For instance,

research conducted by the Trasande group in 2005 pointed out that approximately 300,000–600,000 children in the USA every year have recorded cord blood levels of Hg > 5.8 μ g/L, thus subjecting them to a higher risk of developmental and neurological problems (Trasande, Landrigan, and Schechter 2005). The resultant loss in intelligence leads to reduced economic productivity that continues throughout the lifetime of affected children. This productivity loss was calculated to be \$ 8.7 billion (ranging between \$2.2 and 43.8 billion). A 2012 calculation suggested that the current exposure to methyl mercury (CH₃Hg) in the USA could be associated with total IQ loss among children aging between 0 and 5 years ($n = 25.5$ million) in the USA population, as 1,590,000 IQ points for this period, or IQ points of 264,000 per year (Grandjean et al. 2012). The estimated value of one IQ point is about \$18,000, and an annual economic loss owing to mercury brain drain could be approximated to be about \$5 billion. A study by Bellanger et al. (2013) on the European Union (EU) has estimated that nearly 1.5–2 million children are born annually having CH₃Hg exposures exceeding the safe limit (0.58 μ g/g) and 200,000 children above the maximum levels of 2.5 μ g/g recommended by the WHO. On country-wise analysis, children born in Spain and Portugal were found to be most exposed to methyl mercury, with Hungary being the least exposed. Within the EU,

TABLE 1 | Toxic effects exerted by heavy metals on various organs based on different animal studies.

Animal model	Organ	Negative effects	References
Arsenic			
Wistar rats, Swiss albino mice, Swiss mice	Brain	↑ LPO, MDA, NO, TNF- α , IL-1 β , IL-6, Nos-2, and ROS	Dhar, Kaushal, and Kumar (2018), Metwally et al. (2020) and Foyzun et al. (2022)
Kunming mice	Heart	↑ MDA, ROS, TNF- α , IL-6, Bcl-2, and Caspase-3	Sun et al. (2021)
Balb/c mice, Wistar rats	Liver	↑ AST, ALT, ALP, LPO, Cytochrome-P450, and TBARS	Santra et al. (2000), Bashir et al. (2006), Messarah et al. (2012)
Kunming mice, Balb/c mice, Sprague Dawley rats	Kidney	↑ ROS, Creatinine, and 8-OHdG	Li et al. (2010), García-Sevillano et al. (2014), and Hassani et al. (2015)
ICR mice, Wistar rats	Testicular	↑ MDA, NO, TNF- α , IL-1 β , and AOPP Decreased epididymal sperm counts, lowered activities of 3 β -hydroxysteroid dehydrogenase (HSD) and 17 β -HSD	Chang et al. (2007), Mahajan et al. (2018), and Mehrzadi et al. (2018)
Wistar rats	Lung	↑ IL-1 β , IL-18, PI3K/AKT, and MMP-9	Wang, Zheng, and Zhang (2021)
Lead			
Wistar rats, Sprague Dawley rats	Brain	↑ GFAP, MDA, Caspase-3, Caspase-9, and LPO	Struzyńska et al. (2001), Soltaninejad et al. (2003), and Maiti et al. (2017)
Wistar rats	Heart	↑ Systolic and diastolic pressure	Fioresi et al. (2014)
Wistar albino rats, Sprague Dawley rats	Liver	↑ AST, ALT, ALP, γ -GT, LPO, and α -GPDH	Jarrar and Taib (2012), Mehana, Meki, and Fazili (2012), and Amin et al. (2021)
Sprague Dawley rats, Wistar rats	Kidney	↑ Total protein, LPO, creatinine, and urea	Abdel-Moneim et al. (2014) and Asiwe et al. (2022)
Albino rats, Kunming mic	Testicular	↑ MDA Reduced sperm count, viability, motility, and reduction in testosterone	El-Magd et al. (2017) and Elsheikh et al. (2020)
Guinea pigs	Lung	↑ Total WBCs, eosinophil, neutrophil, and basophil	Boskabaddy and Farkhondeh (2013)
Mercury			
Sprague Dawley rats	Brain	Overactivation of NMDA receptor	Xu et al. (2012)
Sprague Dawley rats	Heart	Myofibrillar necrosis and mitochondrial damage	Arbi et al. (2021)
Wistar rats	Liver	↑ MDA, AST, ALT, and ALP	Raeeszadeh et al. (2021)
Wistar rats	Kidney	↑ MDA, urea, creatinine, and uric acid	Raeeszadeh et al. (2021)
Wistar albino rats, Wistar rats	Testicular	↑ ROS and TBARS Decreased sperm count, motility, and reduction in testosterone	Boujbiha et al. (2009), Moussa et al. (2011), Kalender et al. (2013)
Sprague Dawley rats	Lung	Collapsed alveolar spaces	Naidoo et al. (2019)

Abbreviations: α -GPDH, alpha-glycerophosphate dehydrogenase; γ -GT, gamma-glutamyltransferase; ↑, increase; 8-OHdG, 8-hydroxy-2'-deoxyguanosine; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AOPP, advanced oxidation protein products; AST, aspartate transaminase; Bcl-2, B-cell lymphoma-2; GFAP, glial fibrillary acidic protein; IL-1 β , interleukin-1 beta; IL-6, interleukin-6; IL-18, interleukin-18; LPO, lipid peroxidation; MDA, malondialdehyde; MMP-9, matrix metalloproteinase-9; NMDA, N-methyl-D-aspartate; NO, nitric oxide; Nos-2, nitric oxide synthase-2; PI3K/AKT, phosphatidylinositol-4,5-bisphosphate 3-kinase/protein kinase B; ROS, reactive oxygen species; TBARS, thiobarbituric acid reactive substances; TNF- α , tumour necrosis factor- α ; WBCs, white blood cells.

the conversion of CH_3Hg effects on brain development into IQ points reported that the total benefits of preventing exposure annually are calculated at IQ points > 600,000 per year, resulting in a total economic benefit ranging between 8000 and 9000 million Euro for each year. Miners in the Philippines, Colombia, Indonesia, Guyana, Zimbabwe, Brazil, and Tanzania have been detected with 50 times the mercury levels recommended by the WHO (Budnik and Casteleyn 2019).

2.2 | Drugs

2.2.1 | Paracetamol/Acetaminophen

Paracetamol (PCM) is among the foremost used medicines around the globe and is easily available in most countries without a prescription (Chidiac et al. 2023). In comparison to other analgesic drugs that are accessible over the counter (OTC), paracetamol has a reasonably narrow therapeutic index (Chidiac et al. 2023). The most prevalent and longstanding mechanistic explanation for the antipyretic and analgesic actions of paracetamol involves the inhibition of cyclooxygenase (COX) enzymes in the central nervous system (CNS). However, there is ongoing debate regarding the specific COX variant or isoenzyme targeted by paracetamol, as well as the nature of its molecular interactions with these enzymes (Ayoub 2021). It has been suggested that paracetamol selectively inhibits COX-2 by acting as a reducing agent despite its relatively low potency in inhibiting both COX-1 and COX-2 enzymes in vitro. In vivo, data from COX-1 transgenic mice indicate that paracetamol primarily mediates its analgesic and thermoregulatory actions, including hypothermia and antipyresis, by inhibiting a specific variant of the COX-1 enzyme (Ayoub 2021). PCM toxicity can be encountered following errors in dosing, accidental exploratory intake in children, and intentional self-harm overdose. The drug can cause severe liver toxicity with merely 10g (or 200mg/kg drug for patients weighing less than 50kg) in case of an acute overdose (Table 2) (Chidiac et al. 2023). Repetitive supratherapeutic ingestion of PCM can cause toxicity with daily doses only slightly higher than the maximum recommended therapeutic dose (Acheampong and Thomas 2016). Of late in the USA, PCM has been documented to be the prime cause of acute liver failure (ALF), with 39% of cases reported due to ALF (Ostapowicz et al. 2002; Reddy et al. 2016). Likewise, the drug primarily contributes to ALF cases in countries like Sweden, Australia, the UK, and South America (Devarbhavi et al. 2023). However, in Spain and Germany, drugs such as antibiotics or non-steroidal anti-inflammatory drugs (NSAID) appear to be more common causes of ALF (Escorsell et al. 2007; Hadem et al. 2008). Complimentary/ancient traditional medicines and therapies for tuberculosis have been the foremost factors leading to acute-on-chronic liver failure (ACLF), drug-induced ALF, and drug-induced liver injury (DILI), in regions of Asia, especially India and China (Devarbhavi et al. 2019).

2.2.2 | Diclofenac

Diclofenac is an NSAID, immensely used for the treatment of acute pain and chronic pain linked to various pathological conditions in animals and humans (Lonappan et al. 2016;

Sathishkumar et al. 2020). Like other NSAIDs, diclofenac exerts its effects by inhibiting prostaglandin synthesis through the inhibition of both COX-1 and COX-2 enzymes, showing relative equipotency. However, extensive research has shown that the pharmacological activity of diclofenac extends beyond COX enzyme inhibition, encompassing multimodal and novel mechanisms of action (MOA) in specific contexts (Gan 2010). The drug is utilized by nearly 7.6 million people annually (Szpot, Wachelko, and Zawadzki 2022). Two main causes for its toxicity include anaphylactic shock after its ingestion and acute drug intoxications (Netter et al. 1984). In Europe, severe cases of anaphylactic shock have been reported in 1–3 persons per 10,000, with a 0.65%–2% mortality rate. In Australia and the USA, these rates are further high (Moneret-Vautrin et al. 2005). Klauw, Wilson, and Stricker (1996) have recognized 30 cases of anaphylactic shock associated with diclofenac in published 773 reports on adverse drug reactions. In the Netherlands, studies have shown that amoxicillin, diclofenac, and glafenine were the top three drugs causing hospital admission to drug-induced anaphylaxis (Klauw et al. 1993). The approximate incidence of diclofenac-related anaphylactic shock affects 1 in 10,000–20,000 patients. In Italy, on examination of diclofenac, it was confirmed that the drug is the most common NSAID related to anaphylaxis (Leone et al. 2005). The need to determine diclofenac levels in the blood can also be associated with its ability to induce fatal hepatitis, deadly toxic epidermal necrolysis, Nicolau syndrome, acute autoimmune hemolytic anaemia, rhabdomyolysis, fatal necrotizing fasciitis which can further lead to death (Table 2) (Szpot, Wachelko, and Zawadzki 2022).

2.2.3 | Colistin/Polymyxin E

Polymyxin E or Colistin is a natural antibiotic isolated first from the bacterium *Bacillus polymyxa* subspecies *colistinus* in 1950 (Kumar et al. 2021). Colistin contains L-diaminobutyric acid, which carries a positive charge, facilitating its binding to the negatively charged phosphate groups in lipid A, a key component of lipopolysaccharide (LPS) in Gram-negative bacilli bacteria (Deris et al. 2014). Lipid A plays a vital role in bacterial permeability and exchange with the cell exterior (Velkov et al. 2010). Colistin acts by displacing divalent cations such as magnesium (Mg^{2+}) and calcium (Ca^{2+}) competitively, disrupting the three-dimensional structure of LPS. Subsequently, the hydrophobic acyl fatty acid chains of colistin insert into the outer membrane (OM), leading to the expansion of the OM monolayer. This causes the OM to become permeabilized, allowing colistin to pass through. The interaction of colistin with the inner membrane (IM) phospholipid bilayer destabilises it, disrupting the integrity of the membrane, impairing its ability to maintain cellular contents, and ultimately causing cell lysis (Velkov et al. 2010). In addition to this, colistin binds to lipid A and exerts anti-endotoxin activity, preventing endotoxin-induced shock. Ultimately, colistin solubilizes the bacterial cell membrane, resulting in a bactericidal effect (Falagas, Kasiakou, and Saravolatz 2005). Nevertheless, the therapeutic use of colistin decreased in the decade of 1970 due to its association with nephrotoxicity and emergence and better availability of newer antibiotics with enhanced safety profiles (Alotaibi et al. 2022). Several studies have revealed that colistin is effective in the treatment of multidrug-resistant gram-negative bacteria

TABLE 2 | Toxic effects exerted by drugs on various organs based on different animal studies.

Animal model	Organ	Negative effects	References
Paracetamol/acetaminophen			
Wistar rats, NMRI mice	Brain	↑ Oxidative stress, reactive astrogliosis, and Nrf2/Keap1 transcriptional ratio ↓BDNF	Vigo et al. (2019) Philippot et al. (2022) and Lalert et al. (2023)
Swiss Webster	Heart	↑ COX-2 and iNOS	Okiljević et al. (2024)
Swiss mice, Wistar rats	Liver	↑ MDA, NO ₂ ⁻ , NO ₃ ⁻ , ALT, AST, and ALP	Mladenović et al. (2009), Rašković et al. (2017), and Soliman et al. (2020)
Wistar albino rats, Sprague Dawley rats	Kidney	↑ Serum urea, creatinine, MDA, IL-6, and TNF-α	Canayakin et al. (2016), El-Maddawy and El-Sayed (2018), and Dallak et al. (2022)
Wistar albino rats	Testicular	Reduced sperm count, motility, alive sperm rate, and testosterone level	El-Maddawy and El-Sayed (2018) and Khayyat (2021)
ICR mice	Lung	↑ CYP2E1	Sandoval et al. (2019)
Diclofenac			
Wistar albino rats	Brain	Reduction in thyroid axis hormones and Brain neurotransmitters	Motawi et al. (2019)
Wistar albino rats, C57BL/ mice	Heart	↑ Meteorin, endoglin, and ROS	Dolanbay et al. (2021) and Thai et al. (2023)
Kunming mice, Wistar rats	Liver	↑ ALT, AST, ALP, MDA, TNF-α, IL-1β, TC, TG, LDL-C, VLDL-C, and NO ₂ ⁻	Huang et al. (2016), Nouri et al. (2019), and Heidarian and Nouri (2021))
Wistar rats, Wistar albino rats, Sprague Dawley rats	Kidney	↑ Serum creatinine, urea, BUN, ROS, RNS, TNF-α, IL-1β, and IL-6	Aycan et al. (2018), Jerine and Prince (2018), and Alkuraishy, Al-Gareeb, and Hussien (2019)
Albino rats	Testicular	Reduced sperm count, motility, density, and testosterone level ↑ LPO	Vyas, Purohit, and Ram (2019), Mousa et al. (2020), and Waly et al. (2022)
Colistin/polymyxin E			
Kunming mice, Wistar albino rats	Brain	Altered mitochondrial permeability transition (MPT), membrane potential, and activities of mitochondrial succinate dehydrogenase ↑ Glu, GABA, GABAAR, GABABR, NRI, NR2A, NR2B, MDA and Caspase-3 ↑ Acetylcholinesterase, Butyrylcholinesterase, Monoamine oxidases A and B, Ecto-nucleoside triphosphate diphosphohydrolase and Ecto-5' nucleotidase activities	Dai, Li, and Li (2013), Wang et al. (2016), and Ajiboye (2018)
Kunming mice, Wistar rats	Kidney	↑ BUN, MDA, serum creatinine, 8-OHdG, GGT, and MDA Activation of Caspase-3, 8, and 9	Dai et al. (2014) and Ghliissi et al. (2014)
Sprague Dawley rats	Testicular	↑ Caspase-3 Reduced sperm motility	Aksu et al. (2018) and Aksu, Kandemir, and Küçükler (2021)

Abbreviations: ↓, decrease; ↑, increase; 8-OHdG, 8-hydroxy-2'-deoxyguanosine; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate transaminase; BDNF, brain-derived neurotrophic factor; BUN, blood urea nitrogen; COX-2, cyclooxygenase-2; CYP2E1, cytochrome P450 2E1; GABA, gamma aminobutyric acid; GABAAR, gamma aminobutyric acid type A receptor; GABABR, gamma aminobutyric acid type B receptor; GGT, gamma-glutamyltransferase; Glu, glutamate; IL-1β, interleukin-1 beta; IL-6, interleukin-6; iNOS, inducible nitric oxide synthase; Keap1, kelch-like ECH-associated protein 1; LDL-C, low density lipoprotein cholesterol; LPO, lipid peroxidation; MDA, malondialdehyde; NO₂⁻, nitrite; NO₃⁻, nitrate; NR1, n-methyl-D-aspartate 1 receptor; NR2A, n-methyl-D-aspartate 2A receptor; NR2B, n-methyl-D-aspartate 2B receptor; Nrf2, nuclear factor erythroid 2-related factor 2; RNS, reactive nitrogen species; ROS, reactive oxygen species; TC, total cholesterol; TG, triglyceride; TNF-α, tumour necrosis factor-α; VLDL-C, very low density lipoprotein cholesterol.

(MDR-GNB). Additionally, the mortality and cure rates of 60% and 75%, respectively, have been reported (Alotaibi et al. 2022). Colistin-induced nephrotoxicity is the primary dose-limiting adverse effect associated with its usage. Although the efficacy of colistin has been ascertained for decades, the acute kidney injury (AKI) incidence rate, potential predisposing factors and the time taken for AKI have not been determined yet (Table 2) (Miano et al. 2018; Tsuji et al. 2019). The prevalence of colistin-related AKI has been reported to range between 20% and 60% (Alotaibi et al. 2022). The heterogeneity between these studies could be pertaining to the use of various AKI definitions, the wide range of dosage regimens for colistin, and the inclusion of diverse patient populations (Alotaibi et al. 2022). Studies have revealed that the AKI time-to-incidence in patients undergoing treatment with colistin ranges between 5 to 12 days, and the induced nephrotoxicity is often reversible on modifying the dosing or ceasing the treatment (Alotaibi et al. 2022). Various risk factors for colistin-related AKI have been reported. These include prolonged colistin therapy duration, nephrotoxic agent coadministration, advanced age, underlying disease severity, dose-independent higher weight, baseline renal function, higher Charlson Comorbidity index score (CCI score), hypoalbuminemia, chronic comorbidities, involving diabetes mellitus, chronic liver disease, hypertension, and chronic obstructive pulmonary disease (Alotaibi et al. 2022).

2.3 | Carbon Tetrachloride (CCl₄)

CCl₄ is a xenobiotic found in the environment and is known for its hepatotoxic effects in both humans and animals. It is primarily released into the environment through the residual discharge

from chlorofluorocarbon manufacturing units (Garcia-Manieri et al. 2022). Other sources of CCl₄ include cleaning fluids and fire extinguishers (Garcia-Manieri et al. 2022). Upon entering the body, CCl₄ is metabolised by the CYP450 enzyme complex located in the liver's endoplasmic reticulum, resulting in the formation of a highly reactive metabolite, the trichloromethyl radical (CCl₃·). This radical then binds to proteins, lipids, and DNA via alkylation reactions (Pereira, Barros, and Ferreira 2016; Dassarma et al. 2018; McGill and Jaeschke 2019). The alkylation of biomolecules leads to mitochondrial damage, oxidative stress, decreased protein synthesis, altered steatosis, and disrupted calcium homeostasis. The CCl₃· radical also reacts with O₂, forming the CCl₃OO· (trichloro methyl peroxy) radical, which initiates lipid peroxidation, further damaging cellular membranes. Several studies have demonstrated that the hepatotoxicity of CCl₄ is primarily driven by two key processes: alkylation and lipid peroxidation (McGill and Jaeschke 2019). Chronic exposure to CCl₄ can lead to liver fibrosis, severe oxidative stress, and cirrhosis, while acute liver injury may occur even after a single dose (Wang, Han, et al. 2020; Wang, Li, et al. 2020; Wang, Yan, et al. 2020). Table 3 summarizes the negative impacts of CCl₄ on various body organs, with a focus on the liver.

2.4 | Acrylamide

Acrylamide is an organic compound, largely produced in the industrial sector for synthesizing polyacrylamide, used as a flocculating agent and dry strength agent in paper making, flocculating agent in wastewater treatment, human soft tissue filler in surgery, oil-displacing agent in the development of oilfield, and anti-erosion soil protective agent especially in cultivated soil

TABLE 3 | Toxic effects exerted by CCl₄ on various organs based on different animal studies.

Animal model	Organ	Negative effects	References
Wistar albino rats, Wistar rats	Brain	↑ MDA, NO, TNF-α, IL-6, TGF-β1, and IL-1β	Ismail, Salem, and Eassawy (2016) and Anadozie et al. (2019)
Wistar albino rats	Heart	↑ CPK, and LDL	Murtaza and Khan (2023)
ICR mice, Balb/c mice, Sprague Dawley rats	Liver	↑ AST, ALT, MDA, TNF-α, IL-1β, γ-GT, TG, CHOP, PERK, H ₂ O ₂ , TBARS, Bax/Bcl-2, and Caspase-3	Sahreen, Khan, and Khan (2013), Li, Wang, and Chen (2017), Lee et al. (2019), Yang, Lin, et al. (2019), Yang, Yin, et al. (2019), and Huang et al. (2020)
Wistar albino rats, NMRI mice, Balb/c mice, Swiss albino mice, Albino rats, Sprague Dawley rats	Kidney	↑ TBARS, H ₂ O ₂ , PC, BUN, creatinine, MDA, Urea, IL-1β, IL-2, TNF-α, uric acid, and albumin	Srinivasan et al. (2005), Khan et al. (2010), Kalantari et al. (2018), Safhi (2018), Zangeneh et al. (2018), and Elsawy et al. (2019)
Wistar rats, Wistar albino rats, Sprague Dawley rats	Testicular	↑ Epididymal sperm concentration, MDA, TBARS, Caspase-3, H ₂ O ₂ , and LPO	Sönmez et al. (2014), Yüce et al. (2014), Moneim (2016), Shah and Khan (2017), and Rahmouni, Daoud, and Rebai (2019)
Wistar rats, Sprague Dawley rats	Lung	↑ TBARS, H ₂ O ₂ , NO ₂ ⁻ , TNF-α, IL-1β, IL-2, and NO synthase	Mizuguchi et al. (2006) and Khalil et al. (2020)

Abbreviations: ↑, Increase; γ-GT, gamma-glutamyltransferase; ALT, alanine aminotransferase; AST, aspartate transaminase; Bax/Bcl-2, Bax-Bcl-2 associated X protein; BUN, blood urea nitrogen; CHOP, C/EBP homologous protein; CPK, creatine phosphokinase; H₂O₂, hydrogen peroxide; IL-1β, interleukin-1 beta; IL-2, interleukin-2; IL-6, interleukin-6; LDH, lactate dehydrogenase; LPO, lipid peroxidation; MDA, malondialdehyde; NO, nitric oxide; NO₂⁻, nitrite; PC, protein carbonylation; PERK, PKR-like ER kinase; ROS, reactive oxygen species; TBARS, thiobarbituric acid reactive substances; TGF-β1, transforming growth factor-β 1; TNF-α, tumour necrosis factor-α.

TABLE 4 | Toxic effects exerted by acrylamide on various organs based on different animal studies.

Animal model	Organ	Negative effects	References
Wistar albino rats, Wistar rats	Brain	↑ MDA, LDL, GFAP, and Oxo-8dG	Erdemli et al. (2016), Abdel-Daim et al. (2020), and Farag et al. (2021)
Wistar rats, Wistar albino rats	Heart	↑ MDA, TC, TG, LDL-C, VLDL-C, and NF-κB	Abdel-Moneim et al. (2019) and Coşgun, Coşkun, and Celik (2022)
Sprague Dawley rats, Wistar albino rats	Liver	↑ AST, ALT, ALP, MDA, and Keap 1 protein	Gedik et al. (2017) and Donmez et al. (2020)
Wistar albino rats, Sprague Dawley rats	Kidney	↑ MDA, TNF-α, IL-1β, urea, and creatinine	Bedir et al. (2021) and İbaokurgil et al. (2023)
Sprague Dawley rats, Sprague Dawley albino rats	Testicular	↑ MDA, NO, 38α MAPK, TNF-α, NF-κB, IL-1β, IL-6, COX-2, Cytochrome c, Bax, Caspase-3, LC3A, LC3B, and beclin-1 protein levels	Elghaffar et al. (2015) and Kucukler et al. (2020)
B6C3F1 mice, Balb/c mice	Lung	↑ IgE, IL-4, IL-5, and IL-13 ↓ H4K20me3	Conti et al. (2019) and Hajimohammadi et al. (2020)

Abbreviations: ↑, increase; ↓, decrease; 38α MAPK, p38α mitogen-activated protein kinase; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate transaminase; Bax, Bcl-2 associated X protein; COX-2, cyclooxygenase-2; GFAP, glial fibrillary acidic protein; H4K20me3, H4 lysine 20 trimethylation; IgE, immunoglobulin E; IL-1β, interleukin-1 beta; IL-13, interleukin-13; IL-4, interleukin-4; IL-5, interleukin-5; IL-6, interleukin-6; Keap 1, kelch-like ECH-associated protein 1; LC3A, light chain 3A; LC3B, light chain 3B; LDL, lactate dehydrogenase; LDL-C, low density lipoprotein cholesterol; MDA, malondialdehyde; NF-κB, nuclear factor kappa B; NO, nitric oxide; Oxo-8dG, 8-oxo-2'-deoxyguanosine; TC, total cholesterol; TG, triglyceride; TNF-α, tumour necrosis factor-α; VLDL-C, very low density lipoprotein cholesterol.

TABLE 5 | Toxic effects exerted by alcohol on various organs based on different animal studies.

Animal model	Organ	Negative effects	References
C57BL/6 mice	Brain	↑ Oxo-8dG and TUNEL	Fowler et al. (2014)
Friend Virus B-type mice, Albino Friend Virus B-type mice	Heart	↑ CPK, Bax, and Caspase-3	Kannan, Wang, and Kang (2004) and Guo and Ren (2010)
Sprague Dawley rats, ICR mice	Liver	↑ MDA, LPO, COX-2, TNF-α, IL-6, ALT, AST, ALP, and IL-1β	Kim et al. (2015), Okamura et al. (2018); Khan et al. (2019a), Khan, Salhotra, et al. (2019), Liang, Xi, et al. (2021), and Liang, Yang, et al. (2021)
C57BL/6 mice, Wistar rats	Kidney	↑ Urea, creatinine, TBARS, and NAG Induced mitochondrial protein acetylation and Oxidative stress	Das et al. (2008), Harris et al. (2015), and Costa-Valle et al. (2018)
Sprague Dawley rats, Wistar rats	Testicular	↑ LPO and MDA Lowered activities of 3β-hydroxysteroid dehydrogenase (HSD), and 17β-HSD	Swathy, Panicker, and Indira (2006), Kurus et al. (2009), Priya, Girish, and Reddy (2014), and Akang et al. (2015)
C57BL6/J mice	Lung	↑ MIP-2 and KC	Massey et al. (2015)

Abbreviations: ↑, increase; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate transaminase; Bax, Bcl-2-like protein; COX-2, cyclooxygenase-2; CPK, creatine phosphokinase; IL-1β, interleukin-1 beta; IL-6, interleukin-1; KC, keratinocyte chemoattractant; LPO, lipid peroxidation; MDA, malondialdehyde; MIP-2, macrophage inflammatory protein-2; NAG, n-acetyl-beta-D-glucosaminidase; Oxo-8dG, 8-oxo-2'-deoxyguanosine; TBARS, thiobarbituric acid reactive substances; TNF-α, tumour necrosis factor-α; TUNEL, terminal transferase-mediated biotin dUTP nick end labelling.

for agriculture sector (Fan et al. 2023). However, acrylamide's health effects had been underemphasized for decades until its wide presence was detected in thermally processed starch-rich foods by scholars from Stockholm University and the Swedish National Food Administration (SNFA) in 2002. Their findings revealed the mechanism, wherein it was found that acrylamide formation was mediated at higher temperatures (usually

exceeding 120°C) through the Maillard reaction (Mottram, Wedzicha, and Dodson 2002; Stadler et al. 2002). Since the surfacing of these findings, a constant attempt has been made to bring awareness for acrylamide pollution caused by thermally processed food products and its possible adverse effects on human health, alongside the relevant research conduction including its toxicity assessment on human health, development of

method for acrylamide determination, pollution control and production regularity in the thermal procedure have rapidly gained grave concerns of the scientific community, governments, and organizations (Mottram, Wedzicha, and Dodson 2002; Stadler et al. 2002; Tareke et al. 2002; Timmermann et al. 2021). Acrylamide is a well-known toxic compound possessing potential carcinogenicity, genetic toxicity, neurotoxicity, and developmental and reproductive toxicity with higher permeability (Table 4). Acrylamide can enter the human body directly through the digestive tract, skin, respiratory tract, and other means; hence, the health risks associated with acrylamide should be taken up with gravity (Mesías and Morales 2016). The exposure to dietary acrylamide has resulted in nearly 23,688.09 DALYs for cancers amongst the Chinese population, as reported in 2016, accounting for 1331.93 DALYs, 243.32 DALYs, 248.91 DALYs, and 176.28 DALYs for breast cancer, endometrial cancer, ovarian cancer, and kidney cancer, respectively (Li, Liu, Wang, et al. 2022; Li, Liu, Xia, et al. 2022). A prospective study by NewGeneris European on mother–child was conducted across Europe wherein 1101 pairs of mother–child were assigned. It was found that mothers fed on an acrylamide-rich diet had greater levels of glycidamide (GA)-haemoglobin (Hb) and acrylamide-Hb adducts in the umbilical cord blood, also indicating a negative association between adduct levels and head circumference and birth weight (Pedersen et al. 2012).

2.5 | Alcohol

Current data suggests that roughly 43% of people across the world indulge in alcohol drinking. Excessive alcohol intake or drinking presents a considerable health risk, resulting in illnesses, disabilities, and even mortality around the world (Devarbhavi et al. 2023). Acute intoxication is usually the limit to end ongoing consumption (Birková et al. 2021). Besides acute intoxications, which often require immediate attention and care, especially among young people, the hazard of alcohol consumption remains primarily with alcohol addiction, resulting in chronic abuse and damage to organs (Birková et al. 2021). Affected systems/organs include the cardiovascular system, the liver, the endocrine system, the neural system, the gastrointestinal tract or the metabolism of the basic nutrients (Table 5) (Birková et al. 2021). Although the systems/organs mentioned above perform diverse functions, the distribution and metabolism of ethanol are variable in these systems/organs. The 2016 data reflects upon an annual alcohol intake of 6.4 L for a person 15 years of age or older worldwide (Rossow and Mäkelä 2021). The global incidence of alcohol use disorder (AUD) has been reported to be 5.1%, amounting to about 283 million people. AUD has the highest incidence in Europe, wherein the affected population constitutes 3.5% of women and 14.8% of men, and in the USA, 5.1% of women and 11.5% of men have the condition. It is worth mentioning that the prevalence of AUD has been reported to be greater in high-income nations compared to low-income nations, where there is a fair possibility of cases getting under-reported and many left undiagnosed (Devarbhavi et al. 2023). Age-standardized proportions of heavy drinking have indicated the most cases in Europe, whereas Asian countries are more likely to bear a greater absolute burden (Singh et al. 2019). The age group of the population falling between 15 to 19 years has more than 40% (highest) of individuals involved in heavy

alcohol drinking, with the most prevalence in Gabon, Congo, Czechia, Equatorial Guinea, Estonia, Ireland, Latvia, Slovenia, and Lithuania. Alcoholism enhances the susceptibility to develop liver disorder-related, cardiovascular and cancer mortality by 260 times, 3.2 times, and 5.1 times, respectively (Hagström et al. 2021). Alcohol consumption has been the predominant cause of hepatic cirrhosis globally, with roughly 60% of cases of cirrhosis being reported in Latin America, North America, and Europe, and approximately 35% of AUD patients developing a number of alcohol-related liver diseases (ALD) (Stein et al. 2016; Avila et al. 2020). Indeed, the higher rates of heavy alcoholism among populations can directly be linked to a higher incidence of cirrhosis within that country or region (Stein et al. 2016). The most recent data has suggested that the preceding decade had witnessed an aggregate increment in ALD-related deaths in developed nations, including Europe, Asia, the USA, and Latin America (Devarbhavi et al. 2023). Additionally, a noticeable acceleration in alcohol-associated deaths has been reported during the COVID-19 pandemic period. However, the accurate impact of the pandemic on ALD cases yet remains un-established (White et al. 2022).

2.6 | Plastics

The plastic industry tends to have an important impact on the contemporary world and has acquired an essential place in our day-to-day lives. Hence, plastic production has increased swiftly over the last few decades, consequently leading us towards the “Plastic Age,” producing plastic amounting to almost 400 million tons every year, and with an expected doubling of this number by 2050, considering the continuation of current trends (Mansuri et al. 2023). Bisphenol A (BPA) and phthalates (PAEs) are the most common chemicals that serve as raw materials for producing plastics that have an array of applications. Phthalates constitute a group of several chemical compounds that, in addition to plastics, make them more durable and flexible (Neves et al. 2023). These can be frequently present in various products, such as vinyl flooring, medical devices, and food packaging, along with personal care products, including shampoos, fragrances, and lotions (Dueñas-Moreno et al. 2022). Another chemical compound used in plastic production is BPA, mainly employed for epoxy resins and polycarbonate plastics (Dueñas-Moreno et al. 2022). Polycarbonate plastics are utilized in a number of products, including food storage containers, water bottles, baby bottles, and toys (Olivier et al. 2023). The extensive employment and applications of BPA and phthalates have led to widespread human exposure followed by human health hazards (Basso, de Araujo-Ramos, and Martino-Andrade 2022; Wang, Liu, Liu, et al. (2022), Wang, Liu, Zhang, et al. (2022), Wang, Liu, Zhou, et al. (2022), Wang, Nag, et al. (2022), Wang, Wang, et al. (2022)). BPA and phthalates have been recognized as endocrine-disruptive chemicals (EDCs) capable of causing substantial harm to human populations (Moreno et al. 2020; Monisha et al. 2022; Ren et al. 2022) (Table 6).

Several reports from Italy, Iran, China, Brazil, Mexico, and India have recorded exposure to phthalates and BPA at levels above the recommended limits or “safe” exposure doses by regulatory agencies (Dueñas-Moreno et al. 2023). Gao et al. (2023) have reported that among Chinese teenagers and children, the dietary

TABLE 6 | Toxic effects exerted by plastics on various organs based on different animal studies.

Animal model	Organ	Negative effects	References
Bisphenol			
Kunming mice, C57BL/6J mice	Brain	↑ Oxidative stress ↓ Neuron numbers, CNPase, MBP, and MAG-1	Zhou et al. (2017), Khan, Bhardwaj, et al. (2019), and Khan, Salhotra, et al. (2019)
Wistar rats, Wistar albino rats	Heart	↑ Caspase-3, and MDA Myocardial fibrosis	Bahey, Abd Elaziz, and Gadalla (2019) and Badawy et al. (2022)
Wistar albino rats, Wistar rats	Liver	↑ AST, ALT, ALP, and total bilirubin	Hassan et al. (2012) and Kazemi et al. (2017)
Wistar rats	Kidney	↑ Caspase-3, 8, 9, and 10, ROS, RNS, MDA, H ₂ O ₂ , TNF-α, and IL-1β	Jiang et al. (2020)
Wistar rats	Testicular	↑ MDA	Kalender, Apaydin, and Kalender (2019)
Wistar albino rats, Wistar rats	Lung	↑ IL-1β, IL-6, MDA, NO, 8-OHdG, Caspase-3, and TNF-α	Faheem, El Askary, and Gharib (2021) and Rehman et al. (2021)
Phthalate			
Specific pathogen-free Sprague Dawley rats	Brain	↑ TUNEL, Caspase-3, AROM ↓ ER-β, BDNF, and p-CREB	Li, Chen, et al. (2013), Li, Jiang, et al. (2013), and Li et al. (2014)
ICR mice	Heart	↑ Oxidative stress	Shen et al. (2023)
Sprague Dawley rats	Liver	↑ α-SMA, COL-I, COL-III, and TGF-β1	Zhao et al. (2019)
Wistar rats, Specific pathogen-free ICR mice	Kidney	↑ ROS, MDA, TNF-α and IL-6	Ashari et al. (2020) and Gu et al. (2021)
Sprague Dawley rats, Wistar rats	Testicular	Atrophy due to apoptosis and necrosis Reduction in sperm count and motility	Park, Habeebu, and Klaassen (2002) and El-Beshbishy et al. (2014)
Wistar rats, C57BL/6J mice	Lung	↓ Nrf2, NQO1, and HO-1 ↑ TSLP and Neutrophil	Wang, Li, et al. (2020), Wang et al. (2020), Wang, Yan, et al. (2020), Wang, Liu, et al. (2022), Wang, Liu, Zhang, et al. (2022), Wang, Liu, Zhou, et al. (2022), and Wang, Nag, et al. (2022), Wang, Wang, et al. (2022)

Abbreviations: α-SMA, alpha-smooth muscle actin; ↑, increase; ↓, decrease; ↓, inhibit; 8-OHdG, 8-hydroxy-2'-deoxyguanosine; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AROM, aromatase cytochrome P-450; AST, aspartate transaminase; BDNF, brain-derived neurotrophic factor; CNPase-2',3', cyclic nucleotide 3' phosphodiesterase; COL-I, collagen I; COL-III, collagen-III; CREB, c-AMP-responsive element binding protein; ER-β, estrogen receptor-β; H₂O₂, hydrogen peroxide; HO-1, heme oxygenase-1; IL-1β, interleukin-1 beta; IL-6, interleukin-6; MAG-1, myelin-associated glycoprotein-1; MBP, myelin basic protein; MDA, malondialdehyde; NQO1, NAD(P)H quinone oxidoreductase 1; Nrf2, nuclear factor erythroid-2-related factor 2; RNS, reactive nitrogen species; ROS, reactive oxygen species; TGF-β1, transforming growth factor-β1; TNF-α, tumour necrosis factor-α; TSLP, thymic stromal lymphopoietin; TUNEL, terminal transferase-mediated biotin dUTP nick end labelling.

intake of di(2-ethylhexyl) phthalate (DEHP) accounted for 65%–70% of daily exposure, and subsequent nondietary ingestion (mainly dust), attributing to 26%–30% of the exposure. The authors also noticed that the data collected from 2010 to 2022 showed a decrease in DEHP exposure with an increase in age amongst the Chinese population. Specifically, school children in the age range of 6–<9 years (both female and male) showed higher susceptibility towards DEHP exposure, wherein 30% of the data collected from the subset surpassed the safe limit recommended by the independent organization, the United States

Environmental Protection Agency (U.S. EPA) (20 μg/kg/bw/d). Even though human phthalates exposure across all age groups has several times higher magnitude than BPA, with greater phthalates exposure to children than adults, the worldwide trend for exposure to BPA through ingestion (nondietary and dietary) has been contrary to the phthalate exposure. On the whole, the values for exposure daily intake (EDI) for BPA and phthalates reported worldwide ranged between 3.00×10^{-5} – $6.56 \mu\text{g/kg/bw/d}$ and 1.11×10^{-7} – $3700 \mu\text{g/kg/bw/d}$, respectively (Dueñas-Moreno et al. 2022). All the same, an increase in EDI value

(rising to 5100 µg/kg/bw/d) for phthalates associated with dose-additive effect has been indicated, thus representing a higher risk with respect to carcinogenic (CR) and noncarcinogenic (HQ) effects (Dueñas-Moreno et al. 2023). The global HQ values of BPA and phthalates ranged from 2.74×10^{-7} – 9.72×10^{-2} and 2.25×10^{-7} –3.66, respectively (Dueñas-Moreno et al. 2023).

2.7 | Nicotine

Nicotine is a natural alkaloid isolated from the dried stems and leaves of *Nicotiana tabacum*, the tobacco plant native to South and North America (Alkam and Nabeshima 2019). Tobacco was first used by Native Americans (for instance, the Anishinaabe-Ojibwe inhabitants living around the Great Lakes region of Canada and America) in several different ways, including religious, medicinal and ceremonial practices millennia ago and have continued the tradition up till now (Struthers and Hodge 2004). Chemically, nicotine is 3-(1-methyl-2-pyrrolidinyl)pyridine, a hygroscopic pale yellow oily liquid. Worldwide tobacco smoking accounted for almost 1.1 billion people in 2015, with more than 50% residing in eight countries or regions—Bangladesh, Brazil, the European Union, China, India, the Russian Federation, the United States, and Indonesia (Alkam and Nabeshima 2019). As per 2020 data, nearly 22.3% of tobacco was consumed by people around the world, with 7.8% women and 36.7% men (WHO 2023). Tobacco is consumed in various forms, but the most prevalent is cigarettes, with an estimated global sale of cigarettes amounting to 5.9 trillion in 2006 (Alkam and Nabeshima 2019). Depending upon this data, it can be estimated that a smoker, on average, smokes 15 cigarettes in a day. A single cigarette contains nicotine varying between 8 and 20 mg, and on average, it is 12 mg in one cigarette; it has been documented that merely >40 mg of nicotine ingestion at a time is ample for killing an adult (Mayer 2014). On calculation, an estimated lethal dose for nicotine in adults has been documented between 40 and

60 mg (0.8–1 mg/kg) and for children, it has been about 10 mg (1 mg/kg) (Alkam and Nabeshima 2019). The severity of intoxication by nicotine depends upon the dose, its frequency and duration, nicotine formulation, interpersonal variability, and route of exposure (Alkam and Nabeshima 2019). The time taken by nicotine to accumulate in the brain alongside other body organs, with consequent pharmacologic effects, is completely dependent upon the rate and route of dosing (Table 7) (Benowitz, Hukkanen, and Jacob III 2009).

2.8 | Mycotoxins

2.8.1 | Aflatoxins

Aflatoxins (AFs) are a group of mycotoxins produced as metabolites by fungi like *Aspergillus parasiticus* and *Aspergillus flavus*. They are commonly present in soybeans, rice, mouldy corn, peanuts, and other oil and grain crops. A few naturally occurring examples include Aflatoxins G1 (AFG1), G2, B1, B2, and hydroxylated metabolites of aflatoxin M1 (AFM1) and M2, with AFB1 being the most toxic of all (Wang, Liu, Liu, et al. (2022), Wang, Liu, Zhang, et al. (2022), Wang, Liu, Zhou, et al. (2022), Wang, Nag, et al. (2022), Wang, Wang, et al. (2022)). Humans get exposed to aflatoxins by consumption of aflatoxin-contaminated food products or ingestion of animal-based foods, wherein animals had previous exposure to aflatoxins (Leong et al. 2012). Since AFB1 displays toxic, immunotoxic, mutagenic, carcinogenic and teratogenic effects on animals and humans, it has been classified by the International Agency for Research on Cancer (IARC) as a group1 carcinogen substance (Ostry et al. 2017). AFB1 is a potent carcinogen that can affect organs such as kidneys and liver (Li et al. 2018; Alvarez et al. 2020). Table 8 summarizes various toxicological effects exerted by aflatoxins through different mechanisms. In recent times, tropical climate regions, such as Tanzania and Kenya, have witnessed multiple

TABLE 7 | Toxic effects exerted by nicotine on various organs based on different animal studies.

Animal model	Organ	Negative effects	References
Wistar rats, F344 and HIV1Tg rats	Brain	↓ Mito-ETC ↑ NO, TNF-α, IL-1β, and IL-6	Das et al. (2009) and Royal III et al. (2018)
Sprague Dawley rats	Heart	↑ Caspase-3, 9, and ROS	Jia et al. (2020)
Sprague Dawley rats, Balb/c mice	Liver	↑ LPO, AST, ALT, ALP, and NO	El-Sokkary, Cuzzocrea, and Reiter (2007) and Jalili et al. (2015)
Sprague Dawley rats	Kidney	↑ MDA, and NO	Akkoyun and Karadeniz (2016)
BALB/c mice, Wistar rats	Testicular	↑ MDA, NO, and TBARS Decline testosterone level	Mosbah, Yousef, and Mantovani (2015) and Jalili et al. (2021)
C57BL/6 mice	Lungs	↑ Nitrotyrosine protein-adduct (oxidative-stress marker, soluble-fraction) and Ub/p62/VCP (impaired-autophagy marker, insoluble-fraction)	Bodas et al. (2016)

Abbreviations: ↑, increase; ↓, inhibit; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate transaminase; IL-1β, interleukin-1 beta; IL-6, interleukin-6; LDH, lactate dehydrogenase; LPO, lipid peroxidation; MDA, malondialdehyde; Mito-ETC, mitochondrial electron transport chain; NO, nitric oxide; ROS, reactive oxygen species; TBARS, thiobarbituric acid reactive substances; TNF-α, tumour necrosis factor-α.

TABLE 8 | Toxic effects exerted by fungal toxins on various organs based on different animal studies.

Animal model	Organ	Negative effects	References
Aflatoxins			
Wistar rats, Balb/c mice	Brain	↑ LPO ↓ Neuron numbers	Bahey, Abd Elaziz, and Gadalla (2015) and Gugliandolo et al. (2020)
Wistar albino rats	Heart	↑ MDA	Yilmaz et al. (2018) and Altyar et al. (2023)
F344 Fischer rats, Wistar albino rats, Wistar rats	Liver	↓ GH signaling ↑ MDA, NO, 4-HNE, IL-10, ROS, and RNS	Knipstein et al. (2015) and Eraslan et al. (2017)
Wistar albino rats, Wistar rats	Kidney	↑ MDA, NO, 4-HNE, IL-10, ROS, and RNS	Eraslan et al. (2017) and Owumi et al. (2020)
Wistar rats, Sprague Dawley rats	Testicular	↓ Sperm count, motility, and viability	Supriya, Girish, and Reddy (2014) and Omur et al. (2019)
Balb/c mice	Lung	↑ TNF-α	Shao et al. (2019)
Ochratoxins			
C57BL/6 mice	Brain	↓ NSCs	Paradells et al. (2015)
Wistar albino rats, C57 mice	Heart	↑ CPK, LDH, and MDA Cytoplasmic vacuole formation	Okutan, Aydin, and Ozelcelik (2004) and Cui et al. (2020)
Wistar rats, Sprague Dawley rats	Liver	↑ MDA ↓ CX26, CX32, CX43, and E-cadherin	Gagliano et al. (2006) and Sutken et al. (2007)
Wistar rats, C57BL/6 mice	Kidney	↑ Kim-1, COX-2, Lcn2, Clu, α-SMA, and TGF-β	Zhu et al. (2016), Li et al. (2021), and Li, Liu, et al. (2021)
Wistar rats	Testicular	Testicular degeneration, seminiferous tubule atrophy, and dissociation of germinative epithelium	Malekinejad et al. (2011)
C57 mice	Lung	↑ Oxidative stress	Ou et al. (2023)
Zearalenone			
Sprague Dawley rats	Liver	↑ AST, ALT, ALP, MDA, and CYP2E1	Zhou et al. (2015) and Gao et al. (2018)
Sprague Dawley rats, Balb/c mice	Kidney	↑ TLR4, MDA, TNF-α, IL-1β, and IL-6	Jia et al. (2014) and Ben Salem et al. (2015)
Kunming mice	Testicular	↓ Sperm count, semen quality, and serum testosterone concentrations	Yang et al. (2007) and Zhou et al. (2020)

Abbreviations: α-SMA, alpha-smooth muscle actin; ↑, increase; ↓, decrease; 4-HNE, 4-hydroxynonenal; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate transaminase; Clu, clusterin; COX-2, cyclooxygenase-2; CPK, creatine phosphokinase; CX26, connexin26; CX32, connexin32; CX43, connexin43; CYP2E1, cytochrome P450 2E1; GH, growth hormone; IL-10, interleukin-10; IL-1β, interleukin-1 beta; IL-6, interleukin-6; Kim-1, kidney injury molecule-1; Lcn2, lipocalin-2; LDH, lactate dehydrogenase; LPO, lipid peroxidation; MDA, malondialdehyde; NO, nitric oxide; NSCs, neural stem cells; RNS, reactive nitrogen species; ROS, reactive oxygen species; TGF-β, transforming growth factor-β; TNF-α, tumour necrosis factor-α.

acute aflatoxin outbreaks due to its exposure (Awuor et al. 2017; Kamala et al. 2018). Ezekiel et al. (2021) have determined liver cancer risk in households consuming nuts and cereals regularly. The researchers also noticed that in children, the risk for liver cancer associated with AFB1 exposure was 2-fold in adolescents and 6-fold in adults.

In contrast, a study by Nabizadeh et al. (2018) revealed no significant differences in the exposure margin between children and adults of the Zanjan Province in Iran despite being a significant liver cancer risk for both groups and consuming

AFB2-contaminated unrefined olive oil. The suggested possible reason could be associated with lower daily olive oil intake in children. Islam et al. (2021) investigated 62 women's breast milk samples collected from a Bangladesh cohort study. They found that more than 50% of the samples contained AFM1, probably due to mothers' consumption of AFB1-contaminated food. Additionally, aflatoxin exposure of 1.34 ng/kg/bw/d was caused by contaminated *Rastrineobola argentea* or *Omena* consumption in Kenya, as found by Marijani et al. (2020). Another study conducted in Tunisia has revealed daily AFB1 intake from contaminated pearl millet consumption to be 3.89 ng/kg/bw/d.

The researchers concluded that the millet could serve as a potential route for aflatoxin exposure among Tunisian consumers (Lasram, Hamdi, and Ghorbel 2020). Furthermore, it has been reported in Caldas, Colombia, that consumers had an exposure of 3.093 and 0.732 ng/kg/bw/d of AFB1 after consumption of rice and “arepas” or maize tortillas, respectively (Martinez-Miranda et al. 2019). A similar study in Spain has reported an exposure level of AFB1 ranging between 0.17 and 0.37 ng/kg/ bw/d in infants consuming cereal-based baby foods assessed for aflatoxin exposure (Herrera et al. 2019).

3 | Antioxidant Potential of Curcumin Mediated Through Nuclear Factor ERYTHROID 2-Related Factor 2 (NRF2)

Oxidative stress occurs when levels of oxidative molecules surpass the body's antioxidant defenses, disrupting its balance and causing damage to lipids, DNA, and proteins, which play critical roles in various pathological conditions (Ashrafizadeh et al. 2020). In chicken embryonic fibroblast (CEF) cells, heat-induced stress stimulates the generation of reactive oxygen species (ROS) and triggers apoptosis while reducing the activity of key antioxidant enzymes, including superoxide dismutase (SOD), glutathione peroxidase (GPX), and catalase (CAT). Curcumin administration has been shown to counteract these harmful effects by reducing oxidative damage through the activation of the mitogen-activated protein kinases (MAPK)-Nrf2 signaling pathway (Samarghandian, Farkhondeh, and Azimi-Nezhad 2017; Xu et al. 2019). Wafi et al. (2019) showed that curcumin enhances exercise performance, reduces fatigue, and restores force in mice with coronary artery ligation-induced heart failure (HFrEF). The study highlighted curcumin's ability to combat oxidative damage and improve antioxidant defenses by activating the Nrf2 pathway in skeletal muscle. Das and Vinayak (2018) found that curcumin activates enzymes involved in antioxidant defense, reduces pro-inflammatory cytokines expression, and enhances the Nrf2 signaling pathway, particularly through NAD(P)H quinone oxidoreductase-1 (NQO-1) gene activation, in Dalton's lymphoma ascite cells. Additional studies have documented curcumin's ability to regulate various biochemical and physiological processes. Curcumin reduces feed conversion (FC) ratios and suppresses abnormal elevations in serum corticosterone and malondialdehyde (MDA) levels. It also enhances mitochondrial GSH concentrations in the liver and stimulates the expression of phase II metabolic enzyme genes via Nrf2 pathway activation (Wang et al. 2018). Furthermore, curcumin demonstrates significant anti-inflammatory effects by inhibiting tumor necrosis factor- α (TNF- α) and interleukin-1 beta (IL-1 β) production in paw skin and peritoneal cavity. It has been indicated that the antioxidant and anti-inflammatory activities of curcumin result from Nrf2 pathway activation (Fattori et al. 2015). Boyanapalli et al. (2014) established that curcumin's antioxidant and anti-inflammatory effects depend on Nrf2 activation. Their findings showed that curcumin enhances antioxidant enzyme activity and reduces inflammatory gene expression in macrophages with functional Nrf2 (+/+), whereas these effects are absent in macrophages lacking Nrf2 (−/−). Finally, curcumin has been shown to alleviate oxidative damage caused by heat stress, resulting in increased food intake, weight gain, and improved body weight. It reduces MDA levels in serum, liver, and muscle

tissues and lowers inflammatory markers, with its protective effects primarily mediated through Nrf2 pathway activation (Sahin et al. 2012).

4 | Antibacterial and Anti-Inflammatory Activities of Curcumin

In vitro studies have demonstrated that curcumin effectively inhibits the growth of various antibiotic-resistant bacteria, including *Pseudomonas aeruginosa*, *Klebsiella pneumoniae*, and *Acinetobacter baumannii*, isolated from infectious burn wounds. Notably, curcumin exhibits synergistic effects when combined with meropenem, enhancing its antibacterial activity (Xue et al. 2023). Further investigations into gastrointestinal bacteria of bovine and human origin revealed a selective inhibitory effect of curcumin on Firmicutes, such as *Acetoanaerobium* (*Clostridium*) *sticklandii* and *Clostridioides difficile*, while showing no effect on Bacteroidetes, including *Prevotella bryantii* and *Bacteroides fragilis* (Lakes, Richards, and Flythe 2020). Additionally, the conjugation of curcumin with triphenyl phosphonium forms a compound known as Mitocurcumin. This derivative exhibits potent antibacterial activity against pathogens such as *Bacillus subtilis*, *Staphylococcus carnosus*, *Mycobacterium smegmatis*, and *Escherichia coli*. Mitocurcumin also induces morphological changes in *B. subtilis* (Wang et al. 2023).

The pro-inflammatory nuclear factor kappa B (NF- κ B) signalling pathway plays a crucial role in mammalian intracellular activities and is activated by various stimuli, including pro-inflammatory cytokines such as TNF- α . Upon activation, NF- κ B triggers multiple downstream inflammatory pathways, including those involving inducible nitric oxide synthase (iNOS), lipoxigenase (LOX), and COX-2. Curcumin has been shown to effectively suppress NF- κ B activation, thereby down-regulating these downstream pathways and reducing inflammation. Additionally, curcumin inhibits the expression of pro-inflammatory cytokines, such as CXCL1, mediated through the NF- κ B signaling pathway. This suppression not only alleviates inflammation but also holds the potential to reduce tumor metastasis (El-Saadony et al. 2023).

5 | Role of Curcumin in Detoxification Based on In Vivo Studies

Based on findings from various studies, Figure 3 illustrates the molecular mechanisms underlying curcumin's detoxification process.

5.1 | Heavy Metals

Recently, Cengiz et al. (2023) reported that arsenic (As) induced hepatotoxicity in a rat model, resulting in the highest NF- κ B and IL-1 β levels, serving as markers for systemic inflammation initiation. It was also found that curcumin (Cur) caused a statistically significant decrease in IL-1 β and NF- κ B levels in comparison to the arsenic group. The molecular docking analysis revealed that the interaction of curcumin and arsenic with NF- κ B resulted in

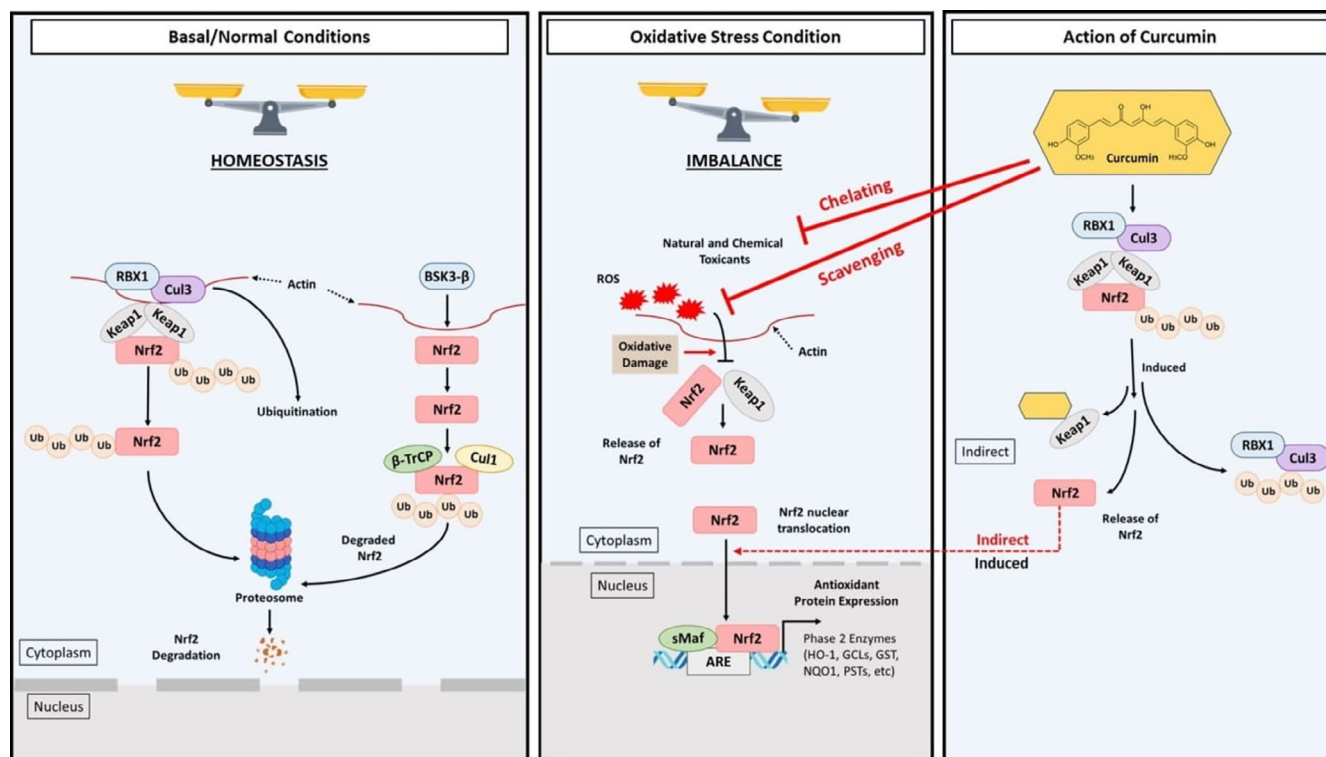


FIGURE 3 | Activation of detoxification signaling pathways by curcumin (Adapted from García-Niño and Pedraza-Chaverri (2014), Licence Number 5925240213008; Tejo and Quintanilla (2021), Creative Commons Attribution (CC BY) license).

the formation of certain complexes affecting the NF- κ B cavity of the active site. The findings of molecular docking indicated that curcumin and arsenic binding takes place mainly at the surface of NF- κ B and active sites. These binding interactions create a steric barrier that impedes substrate entry into the NF- κ B active site, thus limiting molecular orientation inside the active site. This results in interactions that lead to microenvironmental and conformational changes in NF- κ B, thus causing a decreased NF- κ B activity. Based on the *in silico* data, the authors revealed that when arsenic was administered in animal models, a strong reduction in the conventional hydrogen bonding interactions between NF- κ B and arsenic was noticed by the addition of curcumin. Xu et al. (2021) have revealed that curcumin inhibited arsenic-induced kidney and liver toxicity in the Kunming mice. Their findings showed that curcumin exerted both antioxidant and anti-inflammatory action on arsenic-induced kidney and hepatic injury by activation of nuclear factor erythroid 2-related factor (Nrf2) signaling pathways and inhibition of MAPKs/NF- κ B cooperatively. A study by Ishaq et al. (2021) documented that the curcumin alleviation effect was dose-dependent following 240 mg/kg > 160 mg/kg > 80 mg/kg order against arsenic-induced kidney and liver tissues toxicity of rats by diminishing lipid peroxidation, aspartate transaminase (AST), alanine transaminase (ALT), alkaline phosphatase (ALP), urea, total bilirubin, creatinine, total triglycerides, low-density lipoproteins (LDL), and total cholesterol. In addition, curcumin displayed its antioxidant potential by improving its defence system by enhancement in activities of CAT, SOD, glutathione reductase (GR), and GPx (Table 9).

Tubsakul et al. (2021) have investigated curcumin for its effects on oxidative stress, hypertension, and changes in vascular

responsiveness induced by either cadmium (Cd), lead (Pb) or their combined exposure in Sprague–Dawley rats (male). The results have shown that dietary supplementation with curcumin led to a significant reduction in blood pressure, an increase in glutathione and plasma nitrate/nitrite in the blood, and an alleviation of oxidative stress. The effects produced by curcumin could be associated with vascular responsiveness improvement, endothelial nitric oxide synthase upregulation, and nicotinamide adenine dinucleotide phosphate hydrogen (NADPH) oxidase expression downregulation. Furthermore, curcumin decreased metal levels in the aorta, liver, kidney, and blood. A study by Abubakar et al. (2019) investigated curcumin's therapeutic potential in Sprague–Dawley rats (only male) with Pb-induced neurotoxicity. The Pb-administered rats displayed a significant reduction in SOD activity and motor scores with an increase in MDA levels. Additionally, a marked rise in cerebellar lead concentration, along with histological architecture alterations in the layers of the cerebellar cortex, was recorded. On the other hand, curcumin treatment enhanced the motor score, ameliorated oxidative stress markers, reduced cerebellar Pb concentration, and restored cerebellar histological architecture. The study results suggested that curcumin attenuated Pb-induced neurotoxicity through inhibition of chelating activity and oxidative stress.

Li, Mao, et al. (2021) and Li, Lui, et al. (2021) have reported that curcumin administration in Kunming mice ameliorated mercuric chloride (HgCl₂)-associated liver injury by increasing Nrf2 followed by upregulation of heme oxygenase 1 (HO-1) expression under HgCl₂ treatment. In the meantime, HO-1 inhibition by zinc protoporphyrin tends to abolish curcumin cytoprotective effects in HgCl₂-treated hepatocytes. The authors

TABLE 9 | In vivo studies carried out to determine detoxification by curcumin against heavy metals-induced toxicity.

Type of heavy metal	Dose	Animal model	Affected organ	Curcumin dose	Positive effects	References
Arsenic trioxide	2,4, and 8 mg/kg	Albino mice of the Swiss strain	Brain	100 mg/kg	↑ GSH and GPx ↓ MDA	Pandey et al. (2024)
Arsenic trioxide	2,4, and 8 mg/kg	Sanshui white ducks	Brain	400 mg/kg/group	↑ SOD and T-AOC ↓ MDA, TNF- α , IL-1 β , IL-18, IL-2, IL-6, iNOS, and COX-2	Wu et al. (2021)
Sodium arsenite	10, 50, and 100 mg/L	Kunming mice	Liver	200 mg/kg	↑ NQO1, and HO-1 ↓ ALT, AST, and MDA	Gao et al. (2013)
Sodium arsenite	20 mg/kg	Wistar rats	Brain	100 mg/kg	↑ TH	Yadav et al. (2009)
Lead acetate	1%w/v	ICR mice	Brain	100 and 200 mg/kg	↑ AChE ↓ MDA, TNF- α , and COX-2	Changlek et al. (2022)
Lead acetate	50 mg/kg	Wistar rats	Liver	200 mg/kg	↓ ALT, AST, LDH ↓ Oxidative stress, NF- κ B, JNK, and Caspase-3	Alhusaini et al. (2019)
Lead acetate	25 mg/kg	Wistar rats	Brain	30 mg/kg	↓ LPO	Daniel et al. (2004)
Lead acetate	50 mg/kg	Wistar rats	Testicular	100, 200, and 400 mg/kg	↑ SOD, GPx, Sperm motility, Sperm count, and Sperm viability ↓ MDA	Sudjarwo and Sudjarwo (2017)
Methylmercury	5 μ M	Sprague Dawley rats	Brain	0, 2, 5, 10 and 20 μ M	↑ CAT, GSH, NQO1, and HO-1 ↓ LDH and ROS	Yang, Lin, et al. (2019) and Yang, Yin, et al. (2019)
Mercuric chloride	0.6, 1.2, and 2.4 mg/kg	Wistar rats	Liver	100 mg/kg	↑ SOD, GPx ↓ LDH, and ALT	Liu et al. (2017)

Abbreviations: ↑, increase; ↓, decrease; ↑, inhibit; AChE, acetylcholinesterase; ALT, alanine aminotransferase; AST, aspartate transaminase; CAT, catalase; COX-2, cyclooxygenase-2; GPx, glutathione peroxidase; GSH, glutathione; HO-1, heme oxygenase-1; IL-18, interleukin-18; IL-1 β , interleukin-1 beta; IL-2, interleukin-2; IL-6, interleukin-6; iNOS, inducible nitric oxide synthase; JNK-c, jun N-terminal kinase; LDH, lactate dehydrogenase; LPO, lipid peroxidation; MDA, malondialdehyde; NF- κ B, nuclear factor kappa B; NQO1, NAD(P)H quinone oxidoreductase 1; SOD, superoxide dismutase; T-AOC, total antioxidant capacity; TH, tyrosine hydroxylase; TNF- α , tumour necrosis factor- α .

concluded that curcumin could improve Nrf2-mediated HO-1 for upregulation of antioxidant ability, which could be linked to the cytochrome P450 (CYP450) signaling system for suppression of HgCl₂-induced liver damage. Zhao et al. (2021) have studied curcumin's potential protective mechanisms in spleen damage caused by HgCl₂ in male Kunming mice. The data established HgCl₂ induced autophagic cell death, as evidenced by the increase in phosphatidylinositol 3-kinase (PI3K-AKT) protein expression, p62 and microtubule-associated protein 1 light chain 3-II (LC3-II) and the apoptotic cells number. However, curcumin significantly combated sodium overload, calcium leak, and autophagic cell death induced by HgCl₂. Simultaneously, advanced studies have displayed a significant curcumin-mediated activation of the Nrf2 signaling pathway, followed by enhanced antioxidant defences. Overall, activation of the Nrf2 pathway by curcumin administration in animal models to heavy metals has been shown to play a critical role in its detoxification processes.

5.2 | Drugs

Paracetamol (PCM) overdose (3g/kg) induced liver toxicity in Wistar rats by upregulation of TNF- α , interleukin-6 (IL-6), IL-1 β , iNOS, miR-802 gene expression, Cytochrome P450 2E1 (CYP2E1), calcium, and NF- κ B levels (Eassawy, Salem, and Ismail 2021). The study has revealed that oral administration of curcumin exhibited hepatoprotective effects against PCM overdose-induced liver damage in rats through probable regulation of various therapeutic targets such as CYP2E1, NF- κ B, and Nrf2 via organization of miR-802 and miR-122 gene expression. Another study was conducted to evaluate the curcumin effect on the attenuation of PCM-induced mitochondrial changes in CD1 mice liver (Granados-Castro et al. 2016). The findings revealed that dose-dependent curcumin administration prevented PCM-induced hepatic damage. Furthermore, curcumin attenuated the oxygen consumption decrease by use of either succinate or glutamate/malate as substrates (estimated by state 3, uncoupled respiration, adenosine diphosphate/oxygen ratio, and respiratory control ratio) in the synthesis of adenosine triphosphate (ATP), in membrane potential, in the respiratory complexes IV, III and I functioning, and in aconitase activity. Soliman, Nassan, and Ismail (2014) have found that in Sprague Dawley rats, PCM overdose (500 mg/kg) caused an increase in the expression of matrix metalloproteinase-8 (MMP-8) in the coagulative necrosis area surrounding the hepatic lobules' central vein. Curcumin administration also decreased the expression of MMP-8 in the liver of PCM-administered rats.

Recently, Okwakpam et al. (2023) have studied the protective activity of curcumin and NADPH-oxidase inhibitors in diclofenac-induced cardiotoxicity through oxidative stress in Wistar rats (only male). Diclofenac administration significantly ($p < 0.05$) increased the activities of malondialdehyde and NAD(P)H oxidases type 2 while decreasing the activities of CAT, SOD, glutathione (GSH) and GPx levels significantly ($p < 0.05$). Dose-dependent curcumin pre-treatment with 500 mg/kg and 1000 mg/kg of the body weight attenuated all biochemical changes induced by diclofenac. Pre-treatment with curcumin and apocynin, NADPH oxidase 2 (NOX 2) inhibitors were effective in the amelioration of diclofenac-induced cardiotoxicity via oxidative stress alleviation, thus highlighting their

therapeutic potential in the mitigation of diclofenac-induced cardiotoxicity. Ali (2022) has reported that a diclofenac sodium dose of 150 mg/kg induced hepatotoxicity in albino rats (male) by enhancing ALT, AST, nitric oxide (NO), TNF- α , and MDA levels. However, curcumin pre-treatment significantly reduced AST, ALT, NO, TNF- α , and MDA levels and improved SOD levels (Table 10). Curcumin protective potential was studied on colistin-associated peripheral neurotoxicity by using female C57BL/6 mice in a murine model by Dai et al. (2020). Results showed that dose-dependent oral curcumin co-administration brought marked improvement in colistin-induced impaired motor and sensory dysfunctions. Curcumin supplementation at 100 mg/kg and 200 mg/kg caused a significant decrease in lipid peroxidation and upregulation of SOD and CAT activities, Na⁺/K⁺-ATPase activity, and ATP levels in the sciatic nerve tissue in comparison to the only colistin group. Further, supplementation of curcumin at 200 mg/kg also upregulated AKT, mechanistic target of rapamycin (mTOR), nerve growth factor (NGF), HO-1 mRNA, and Nrf2 levels, alongside downregulation of caspases-9 and -3 mRNA activity, and Bcl-2-associated X protein (Bax). Curcumin has demonstrated effective drug-induced toxicity detoxification in a dose-dependent manner.

5.3 | Alcohol

Curcumin exhibited a protective effect on the liver of zebrafish under ethanol-associated (2%) oxidative stress (Song et al. 2022). The study revealed that curcumin treatment caused the down-regulation of genes participating in the Nrf2-Keap1 (Kelch-like ECH-associated protein 1) signaling cascade and modified the expression patterns of genes associated with glutathione metabolism (*gpx1a*, *gsr*, *gstp1*, *idh1a*, and *gsti1*). Treatment with curcumin also enhanced the GSH content and resulted in the recovery of decreased GSH due to ethanol exposure. Ethanol administration at 5 mg/kg triggered apoptosis, neuroinflammation, memory impairment, and reactive oxygen species (ROS), and these effects were significantly inhibited after administration of curcumin, which were assessed by lipid peroxidation (LPO), Nrf2/HO-1 expression, and ROS in the brain of Cb7BL/6 mice under experiment (Ikram et al. 2019). Results indicated that the expression of glial cell markers was regulated by curcumin in ethanol-treated brains of mice, as assessed by the relative TLR4 (Toll-like Receptor 4), GFAP (Glial fibrillary acidic protein), Iba-1 (Ionized calcium-binding adaptor molecule 1), and RAGE (receptor for advanced glycations end products) expression. Also, curcumin downregulated p-NF- κ B (nuclear factor kappa-light-chain-enhancer of activated B cells), p-JNK (Phospho c-Jun N-Terminal Kinase), and downstream target expression. Varatharajulu et al. (2016) have analysed the protective potential of curcumin against alcoholic fatty liver and ethanol-induced liver injury in ω -3 polyunsaturated fatty acids (PUFA) and alcohol (5%) fed Wistar-Furth rats. Curcumin treatment protected against ethanol-induced hepatosteatosis and high ω -3 PUFA. Curcumin upstreamed paraoxonase 1 (PON1) mRNA, thus causing a significant increase in homocysteine thiolactonase and serum PON1 activities in comparison to ethanol and high ω -3 PUFA group. Moreover, curcumin treatment protected against ethanol-associated oxidative stress, indicated by an increase in the antioxidant glutathione levels and a decrease in 4-hydroxynonenal, a lipid peroxidation adduct (Table 11).

TABLE 10 | In vivo studies carried out to determine detoxification by curcumin against drug-induced toxicity.

Type of drug	Dose	Animal model	Affected organ	Curcumin dose	Positive effects	References
Paracetamol	500 mg/kg	Albino rats	Testicular	50 mg/kg	↑ Testosterone, SOD, GPx, Sperm count, and Sperm motility ↓ Oxidative stress and MDA	Mohammed and Sabry (2020)
Paracetamol	200 mg/kg	Wistar albino rats	Liver	100 mg/kg	↓ ALT, AST, and Bcl-2	Ahmad et al. (2019)
Paracetamol	500 mg/kg	Albino rats	Kidney	200, and 400 mg/kg	↑ GSH, and CAT ↓ MDA, Uric acid, Creatinine, and Urea	Ibrahim et al. (2019)
Paracetamol	2 g/kg	Wistar albino rats	Liver, kidney, and testicular	200 mg/kg	↑ GSH, CAT; Testosterone, Sperm count, Sperm motility, and Sperm viability ↓ MDA, AST, ALT, and ALP	El-Maddawy and El-Sayed (2018)
Paracetamol	300 mg/kg	Balb/c mice	Liver	10, and mg/kg	↑ SOD ↓ ALT and MDA	Li, Chen, et al. (2013) and Li, Jiang, et al. (2013)
Paracetamol	650 mg/kg	Wistar rats	Liver, and kidney	50 mg/kg	↑ SOD, CAT, GPx, GST ↓ TBARS, Creatinine, and Urea	Yousef et al. (2010)
Diclofenac	20 mg/kg	Wistar albino rats	Kidney	200 mg/kg	↓ TNF- α , IL-1 β , and KIM-1	Hasan et al. (2023)
Diclofenac	10 mg/kg	Mice	Kidney	100 mg/kg	↑ SOD, and GSH	Ragab et al. (2022)
Diclofenac	100 mg/kg	Albino rats	Kidney	100 mg/kg	↑ CAT, and GSH ↓ MDA, Urea, and Creatinine	Ahmed, Gad, and El-Raouf (2017)
Colistin	300,000 IU/kg	Albino rats	Brain and kidney	200 mg/kg	↑ CAT and GSH ↓ MDA, NO, TNF- α , IL-6, caspase-3, GABA, uric acid, and urea and creatinine	Edrees et al. (2018)

Abbreviations: ↑, increase; ↓, decrease; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate transaminase; Bcl-2, B-cell lymphoma-2; CAT, catalase; GABA, gamma butyric acid; GPx, glutathione peroxidase; GSH, glutathione; GST, glutathione S-transferase; IL-1 β , interleukin-1 beta; IL-6, interleukin-6; KIM-1, kidney molecule-1; MDA, malondialdehyde; NO, nitric oxide; SOD, superoxide dismutase; TBARS, thiobarbituric acid reactive substances; TNF- α , tumour necrosis factor- α .

5.4 | Carbon Tetrachloride (CCl₄)

Peng et al. (2018) have investigated the protective potential of curcumin against CCl₄-associated acute hepatic injury in mice models. Curcumin was administered orally at 50, 100, and 200 mg/kg/day once daily doses for 7 days before exposure to CCl₄. Curcumin pre-treatment at the doses mentioned above significantly reduced CCl₄-associated oxidative stress, indicated by a decrease in MDA formations and an increase

in GSH content and CAT, SOD activities, followed by a reduction in caspase-3 and -9 activities (Table 12). CCl₄-induced inflammation was significantly decreased with curcumin pre-treatment. Furthermore, pre-treatment with curcumin significantly down-regulated the mothers against decapentaplegic homolog 3 (Smad3) mRNAs (both $p < 0.01$) and transforming growth factor beta-1 (TGF- β 1) expression and up-regulated HO-1 and Nrf2 mRNA (both $p < 0.01$) expression in the liver. HO-1 inhibition attenuated the protective action of curcumin

TABLE 11 | In vivo studies carried out to determine detoxification by curcumin against alcohol-induced toxicity.

Alcohol	Dose	Animal model	Affected organ	Curcumin dose	Positive effects	References
Ethanol	10 mL/kg	Sprague Dawley rats	Liver	100, 200, and 400 mg/kg	↑ Nrf2 ↓ AST, ALT, and LDH	Lu et al. (2015)
Ethanol	10 mL/kg	Balb/c mice	Liver	150 mg/kg	↑ HDL-C, SOD, GPx, Nrf2, NQO1, and HO-1 ↓ AST, ALT, TG, TC, and LDL-C	(Xiong et al. (2015)
Ethanol	10 mL/kg	Sprague Dawley rats	Liver	200, and 300 mg/kg	↓ HSCs	Chen et al. (2014)
Ethanol	5 g/kg	ICR mice	Liver	19.7, and 47.5 mg/kg	↑ AMPK ↓ AST, ALT, ALP, and LDH ↓ Fatty acid synthase and Phosphatidate phosphohydrolase	Lee et al. (2013)
Ethanol	0.5, 1, and 2 M	Albino rats	Liver	0.1, 1, and 10 mM	↓ LPO and Cytochrome C release	Ghoneim (2009)

Abbreviations: ↑, increase; ↓, decrease; ↓, inhibit; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AMPK, AMP-activate protein kinase; AST, aspartate transaminase; GPx, glutathione peroxidase; HDL-C, high density lipoprotein cholesterol; HO-1, heme oxygenase-1; HSCs, hepatic stellate cells; LDH, lactate dehydrogenase; LDL-C, low density lipoprotein cholesterol; LPO, lipid peroxidation; NQO1, NAD(P)H quinone oxidoreductase 1; Nrf2, nuclear factor erythroid 2-related factor 2; SOD, superoxide dismutase; TC, total cholesterol; TG, triglyceride.

TABLE 12 | In vivo studies carried out to determine detoxification by curcumin against CCl₄-induced toxicity.

CCl ₄ dose	Animal model	Affected organ	Curcumin dose	Positive effects	References
0.5 mL/kg	Wistar albino rats	Testicular	200 mg/kg	Improvement of testicular morphology ↓ TNF- α and Caspase-3	Bayramova et al. (2024)
0.5 mL/kg	Wistar albino rats	Kidney	200 mg/kg	↓ MDA, LPO, Creatinine, and BUN	Hismiogullari et al. (2015)
0.5 mL/kg	Wistar albino rats	Liver	200 mg/kg	↓ MDA	Hismiogullari et al. (2014)
0.1 mL/100 g	Albino rats	Liver	200 mg/kg	↓ MMP-2	Mohamed and Elbastawisy (2013)
1 mL/kg	Wistar albino rats	Liver	100 mg/kg	↑ SOD, CAT and GPx ↓ AST, ALT, ALP, and TBARS	Gangarapu et al. (2013)
0.2 mL/100 g	Wistar rats	Liver	0.05, 0.5, and 5 g/kg	Activation of CYPs 1A, 2B, 2C and 3A isozymes	Sugiyama et al. (2006)

Abbreviations: ↑, increase; ↓, decrease; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate transaminase; BUN, blood urea nitrogen; CAT, catalase; CYPs, cytochrome P450; GPx, glutathione peroxidase; LPO, lipid peroxidation; MDA, malondialdehyde; MMP-2, matrix metalloproteinase-2; SOD, superoxide dismutase; TBARS, thiobarbituric acid reactive substances; TNF- α , tumour necrosis factor- α .

in case of CCl₄-induced acute hepatic injury. Bayoumi (2017) has also documented that curcumin inhibits TNF- α protein expression, thus ameliorating CCl₄-induced hepatic injury. El-Shafei et al. (2015) have revealed that curcumin treatment also attenuated CCl₄-induced hepatic injury by upregulation of protein kinase B (Akt1) mRNA transcriptional level (the Akt1 signal is pivotal in cell survival, triggered by extracellular matrix and growth factors, thus, showcasing an impaired Akt in liver cirrhosis).

5.5 | Acrylamide

Alaee et al. (2024) have analyzed the curcumin effect on acrylamide (ACR)-induced changes in the ovarian tissue's first-line/primary antioxidant defense in female Balb/c mice. Results have shown that the model group (ACR) had diminished mRNA expression of CAT, SOD, and GPx genes in comparison to the control group. Also, ACR exposure significantly enhanced caspase 3 and Bax expression and diminished expression of the Bcl2l1 gene, leading to a noticeable increase in apoptosis in the ACR group in comparison to that of the control. Curcumin treatment at two different doses (Cur 100 and Cur 200) showed a significant increase in the GPx, CAT, and SOD gene expression and a significant reduction in caspase 3 and Bax expressions, with an increased Bcl2l1 gene expression. Curcumin was administrated at 50 mg/kg and 100 mg/kg oral dose to Sprague–Dawley rats intoxicated with acrylamide, with an ACR exposure at 40 mg/kg for 4 weeks (Guo et al. 2020). A behavioural analysis of all the rats was conducted. ACR-induced motor function deficits, pathological variations in brains, and weight loss showed a significant improvement in 50 mg/kg and 100 mg/kg curcumin-administrated rats. The number of terminal deoxynucleotidyl transferase-mediated dUTP nick-end labelled (TUNEL)-positive apoptotic cells was less in ACR-intoxicated brains treated with curcumin in contrast to the ACR model rat group. Curcumin administration at 100 mg/kg led to upregulation of the expression of telomerase reverse transcriptase (TERT) mRNA and an increase in TERT-positive cell numbers in cortex tissues intoxicated with ACR. Moreover, curcumin treatment diminished the TNF- α , MDA, and IL-1 β concentrations while increasing GSH contents, GSH-Px and SOD activities in the homogenates prepared from the cerebrum of brains, as compared to the ACR control group. A study by Yan et al. (2018) reported that acrylamide exposure of 10 mg/kg/d for 7 weeks caused spatial memory deficits and slight gait abnormality, associated with glial cells activation, phosphorylated cAMP response elements binding protein (P-CREB) reduction and hyperphosphorylated tau aggregation including p-tau (Ser262), PHF1 (p-tau Ser396/404), and AT8 (p-tau Ser202/Thr205) in the Sprague–Dawley rats' cortex and hippocampus. ACR inhibited the lysosomal protease cathepsin D and protein phosphatase 2A (PP2A) to decrease the dephosphorylation and degradation of p-tau. The brain-derived neurotrophic factor (BDNF) and P-CREB showed a significant decrease, as caused by ACR exposure. Upstream signaling of the extracellular signal-related kinase (ERK), Akt, and P-CREB were markedly inhibited. Also, curcumin improved spatial memory impairment induced by ACR through a reversal of tau abnormalities and reduction of P-CREB in the hippocampus.

5.6 | Plastics

Fallahnezhad et al. (2023) have evaluated the protective effect of nanomicellar curcumin (NmCur) in Wistar rats towards bisphenol A (BPA)-induced memory and learning deficits. Results have revealed that BPA upregulated MDA, ROS, TNF- α , IL-1 β , IL-6, SOD, GST, p-JNK, and p-P38 levels and downregulated GPx, GSH, GR, CAT, p-ERK1/2, p-AKT, p-NR1, p-NR2B, p-NR2A, p-GluA1, BDNF and p-CREB levels. BPA reduced step-through latency (STL) along with total and peripheral, but not central, and locomotor activity. It also increased the time taken to locate the hidden platform, distance travelled inside the target quadrant, and mean escape latency time, but at the same, decreased the target quadrant spent time. The combination of NmCur (25 mg/kg and 50 mg/kg) and BPA (50 mg/kg) reversed all the adverse effects produced by BPA (Table 13). Elswefy et al. (2020) have reported that curcumin ameliorated BPA-induced hepatic fibrosis and damage in Wistar rats (male only) mediated through its anti-apoptotic effects that increase the B-cell lymphoma 2 (Bcl-2) immunoexpression in hepatocytes and decrease the hepatic caspase-3 content. Ultimately, curcumin alleviated turnover in the extracellular matrix by downregulating MMP-9 and upregulating tissue inhibitors of MMP-2 gene expression. In vitro and in vivo exposure to BPA-induced differentiation and proliferation of oligodendrocyte progenitor cells (OPCs) was significantly improved by curcumin (Tandon et al. 2020). Curcumin enhanced the protein levels and mRNA expression of myelination markers in rat hippocampus treated with BPA. Curcumin also improved myelination potential through an increase in β -III tubulin-/MBP⁺ cells (neuron-oligodendrocyte co-culture) as well as augmentation of neurofilament/MBP⁺ neurons and fluoromyelin intensity in vivo. The *in silico* docking analysis suggested that Notch pathway genes (Hes-1, Mib-1, and Notch-1) could serve as potential targets for curcumin and BPA. Curcumin reversed the BPA-induced myelination inhibition by increasing the gene expression of the Notch pathway.

Recently, it has been found that di-n-butyl phthalate (DBP) can induce testicular tissue ferroptosis in a dose-dependent manner by upregulation of SP1 expression, which further upregulated direct transcriptional PRDX6, a ferroptosis negative regulator (Bu et al. 2023). The addition of SP1 agonist or overexpression of PRDX6 could help curcumin suppress the testicular ferroptosis induced by DBP. Bu et al. (2023) administered 500 mg/kg/day oral dose of DBP for 3 weeks to Sprague Dawley rats, and elevated testicular tissue ferroptosis levels were detected. When the 300 mg/kg/day dose of curcumin intragastrically and the dose mentioned above of DBP were administered simultaneously, alleviation in the DBP-induced testicular ferroptosis was observed. The findings showed that curcumin was capable of strengthening the self-protection mechanism along with the weakening of ferroptosis levels induced by DBP. Another study by Liang, Xi, et al. (2021) and Liang, Yang, et al. (2021) reported that curcumin exhibited a protective effect on DBP-exposed kidneys in KM mice through blockage of DBP-induced free radicals' formation. Conversely, the curcumin protective effect might also be attributed to the inhibition of p-ERK1/2, Caspase-3, and Bax's high expression in the apoptosis mechanism.

TABLE 13 | In vivo studies carried out to determine detoxification by curcumin against plastic-induced toxicity.

Types of plastic	Dose	Animal model	Affected organ	Curcumin dose	Positive effects	References
Bisphenol A	65 mg/kg	Wistar rats	Liver	100 mg/kg	↑ GSH ↓ MDA	Elijah et al. (2024)
Bisphenol AF	200 µg/L	Zebrafish	Brain	200, 600, and 1200 µg/L	↑ AChE	(Yang et al. (2023)
Bisphenol A	0.5 µg/kg	Albino rats	Stomach	200 mg/kg	↑ Bcl-2 ↓ PCNA	Ismail and El-Meligy (2022)
Bisphenol A	130 mg/kg	Wistar albino rats	Kidney	100 mg/kg	↑ CAT, SOD, GPx, and GST	Aslanturk and Uzunhisarcikli (2020)
Bisphenol A	100 mg/kg	Rats	Testicular	100 mg/kg	↑ Johnson's scoring of tubules	Toor, Irshad, and Asif (2019)
Bisphenol A	130 mg/kg	Wistar rats	Liver	100 mg/kg	↑ CAT, SOD, GPx, and GST	Uzunhisarcikli and Aslanturk (2019)
Dibutyl phthalate	50 mg/kg	Balb/c mice	Spleen	2.5 mg/kg	↑ GSH, and T-AOC ↓ TNF-α, IL-1β, and NF-κB	Wang, Han, et al. (2020), Wang, Li, et al. (2020), and Wang, Yan, et al. (2020)
Di-n-butylphthalate	2 g/kg	Wistar rats	Testicular	200 mg/kg	↑ SOD, CAT, GSH, Testosterone, Sper count, and Sperm motility ↓ MDA	Farombi et al. (2007)

Abbreviations: ↑, increase; ↓, decrease; AChE, acetylcholinesterase; BCL2, B-cell lymphoma-2; CAT, catalase; GPx, glutathione peroxidase; GSH, glutathione; GST, glutathione S-transferase; IL-1β, interleukin-1 beta; MDA, malondialdehyde; NF-κB, nuclear factor kappa B; PCNA, proliferating cell nuclear antigen; SOD, superoxide dismutase; T-AOC, total antioxidant capacity; TNF-α, tumour necrosis factor-α.

5.7 | Nicotine

The therapeutic potential of curcumin in case of nicotine injury in rat hippocampus CA1 region was analysed by Jalili et al. (2019). Results showed that administration of nicotine significantly enhanced the nitrite oxide level as well as decreased total antioxidant capacity along with a decrease in the neurons and neuronal dendritic spines number in comparison to the control group ($p < 0.01$). The study also found that in all nicotine + curcumin groups and curcumin groups, the number of neuronal dendritic spines, total antioxidant capacity, and neurons increased significantly as against the nicotine control ones, while the nitrite oxide level diminished significantly as against the nicotine control ones ($p < 0.01$). Motaghinejad et al. (2017) have assessed CREB–BDNF signaling significance in the mediation of the neuroprotective actions of curcumin in nicotine-induced oxidative stress, inflammation, and apoptosis in rats. The study documented that nicotine perturbed the open field test (OFT) motor activity, and immediate curcumin treatment at 40 mg/kg and 60 mg/kg doses reduced the motor activity disturbances induced by nicotine. Additionally, nicotine treatment enhanced lipid peroxidation as well as the levels of IL-1 β , TNF- α , Bax, and GSH while reducing P-CREB, BDNF, and Bcl-2 levels in the hippocampal region. Also, curcumin tends to activate the P-CREB/BDNF signaling pathway and provides neuroprotection against inflammation induced by nicotine, oxidative stress and apoptosis. Jalili et al. (2014) have shown that curcumin improved the testosterone level, testis weight, sperm motility and count in a dose-dependent manner in nicotine-associated reproductive damage in male mice.

5.8 | Mycotoxins

Zhang et al. (2024) have found that curcumin alleviated the AFB1-associated oxidative stress in the liver and ileum of broilers, indicated by a significant increase in the levels of Keap-1, HO-1, NQO-1, and Nrf2 gene expression ($p < 0.05$) (Table 14). On account of intestinal health, the addition of curcumin led to a significant increase in the ileum relative length ($p < 0.05$), significant elevation in the ileal villi height ($p < 0.05$), a significant reduction in diamine oxidase (DAO) and D-Lactate (D-LA) activities in the serum of broiler ($p < 0.05$). Curcumin alleviated AFB1-induced liver fibrosis and pyroptosis in the ducks via inhibition of the NLRP3 inflammasome and JAK2/STAT3 pathway activation (Cui et al. 2020). Li, Liu, Wang, et al. (2022) and Li, Liu, Xia, et al. (2022) have revealed that supplementation of curcumin in broilers enhanced the damage induced by AFB1 to the immune organs, including the spleen, bursa of Fabricius, and thymus via cytokines expression (IFN- γ) upregulation.

Zhao, Feng, et al. (2024) and Zhao, Jiang, et al. (2024) have studied the effect of dietary curcumin in alleviating the ochratoxin A (OTA) diet-induced damage to juvenile *Ctenopharyngodon idella* or grass carp fish gill barrier. The study found that supplementation of curcumin could significantly offset OTA-induced damage caused to the fish gill barrier mediated through distinct mechanisms: (1) OTA absorption inhibition along with upholding expression of efflux transporters mRNA, thereby reducing OTA residuals; (2) Improving the physical barrier via fortification of antioxidant defenses, strengthening tight junctions,

and reduction of apoptosis; (3) Enhancing the chemical barrier through elevation of goblet cells, antimicrobial peptides, and immune components; (4) Enhancement of the immune barrier via suppression of NF- κ B and TLR4 signaling pathways, resulting in reduced pro-inflammatory factors expression and increase in genes related to T cell differentiation. Curcumin also alleviated muscle toxicity induced by OTA in *C. idella* or grass carp by improving the proteins related to myoblast differentiation (MyoG and MYHC) and myofibrillar development activities, both previously inhibited by OTA (Zhao, Feng, et al. 2024; Zhao, Jiang, et al. 2024). Curcumin could also downregulate the expression of genes related to OTA-augmented protein degradation (*foxo3a*, *ub*, and *murfi*), further related to FoxO3a signaling pathway inhibition. Jiang et al. (2023) have reported that curcumin reduced liver inflammation induced by OTA in ducks through inhibition of the TLR4 signaling pathway as well as regulation of the intestinal microbiota (especially *Bacteroides*). Curcumin plays a crucial role in detoxifying mycotoxins by effectively inhibiting the FoxO3a and NF- κ B signaling pathways.

6 | Curcumin Metabolism and Bioavailability Concerns

In the phase I drug metabolism of curcumin in epithelium cells, it undergoes NADPH-reductase catalyzed reduction, yielding phase I metabolites including octahydro curcumin, hexahydro curcumin, tetrahydro curcumin, and dihydro curcumin; alkaline hydrolysis reaction producing minor hydrolytic products such as ferulic acid, vanillin, feruloyl methane and ferulaldehyde, and autooxidation reaction yielding dicyclopentadiene (Figure 4) (Luis et al. 2017). The phase II metabolism of curcumin involves conjugation reactions that undergo in the presence of conjugative enzymes (catalysts) mostly found in the kidney, intestinal mucosa, and liver. This process paves the way for further metabolism of curcumin and its metabolites generated in the first phase, thus forming more water soluble and bulkier sulfate and glucuronide conjugates (Figure 4). A recent study has revealed that the thus formed conjugates had less activation in human dermal lymphatic microvascular endothelial cells (HMVEC-dLy) in contrast to curcumin owing to their hydrophilicity (Wei et al. 2023). Several clinical and animal studies have also identified these metabolites as the major form of curcumin in serum. These metabolites are primarily excreted out of the body through urine and feces, indicating successful curcumin bioactivities deactivation mediated through phase II metabolic processes and also the rapid curcumin excretion resulting in its poor bioavailability (Lao et al. 2006).

To date, available in vivo and in vitro evidence indicates that only trace amounts of free curcumin reach peripheral blood following oral ingestion, even at high doses (Metzler et al. 2013). After oral intake, the gastrointestinal tract is exposed to higher concentrations of curcumin. However, clinical studies have shown that free curcumin levels in the colorectal tissue of patients receiving up to 4 g of curcumin daily for several days reach only up to 20 nmol/g (Garcea et al. 2005; Carroll et al. 2011). Curcumin has demonstrated cancer-preventive effects in the stomach, colon, and intestine, as well as in extraintestinal regions such as the tongue, soft palate, skin, breast, and sebaceous glands in rodent models (Sharma, Gescher, and

TABLE 14 | In vivo studies carried out to determine detoxification by curcumin against mycotoxins-induced toxicity.

Types of mycotoxins	Dose	Animal model	Affected organ	Curcumin dose	Positive effects	References
Aflatoxin B1	1 mg/kg	Arbor Acres broiler	Liver	300 mg/kg	↑ DNMTs ↓ CYP450	Liu et al. (2023)
Aflatoxin B1	750 µg/kg	Balb/c mice	Kidney	100, and 200 mg/kg	↑ Bcl-2 ↓ Cyt-c, Bax, and Caspase-3, 9	Wang, Liu, et al. (2022), Wang, Liu, Zhang, et al. (2022), Wang, Liu, Zhou, et al. (2022), Wang, Nag, et al. (2022), Wang, Wang, et al. (2022)
Aflatoxin B1	0.75 mg/kg	Balb/c mice	Liver	100 and 200 mg/kg	↑ Nrf2, SOD, CAT, NQO1, and HO-1	Wang, Liu, et al. (2022), Wang, Liu, Zhang, et al. (2022), Wang, Liu, Zhou, et al. (2022), Wang, Nag, et al. (2022), and Wang, Wang, et al. (2022)
Aflatoxin B1	1 mg/kg	Arbor Acres broiler	Liver	300 mg/kg	↓ TNF-α, IL-1β, IL-6, and iNOS	Li, Liu, et al. (2022) and Li, Lui, Xia, et al. (2022)
Aflatoxin B1	60 µg/kg	Duck	Liver	500 mg/kg	↓ H ₂ O ₂ , MDA, and Caspase-3	Jin et al. (2021)
Aflatoxin B1	5 mg/kg	Arbor Acres broiler	Liver	300 mg/kg	↓ ROS, and 8-OHdG	Li et al. (2019)
Aflatoxin B1	5 mg/kg	Broiler chicken	Liver	300, and 450 mg/kg	Activation of Beclin-1, ATG5, Dynein, LC3a, and LC3b-I/II	(Muhammad et al. 2018)
Ochratoxin A	1 mg/kg	Cockerels' broiler	Liver and Kidney	400 mg/kg	↑ GPx4, and Nrf2	Kövesi et al. (2024)
Ochratoxin A	0.5 mg/kg	Sprague Dawley rats	Liver	100 mg/kg	↑ SOD, CAT, and GPx ↓ AST, ALT, ALP, and MDA	Damiano et al. (2021)
Ochratoxin A	0.5 mg/kg	Sprague Dawley rats	Kidney	100 mg/kg	↓ BUN and Creatinine	Damiano et al. (2020)
Zearalenone	40 mg/kg	Balb/c mice	Liver	200 mg/kg	↓ ALT, AST, ALP, ROS, MDA, and H ₂ O ₂	Yang et al. (2024)

Abbreviations: ↑, increase; ↓, decrease; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate transaminase; ATG5, autophagy-related gene 5; Bax, Bcl-2 associated X protein; BCL2, B-cell lymphoma-2; BUN, blood urea nitrogen; CAT, catalase; CYP450, cytochrome P450; Cyt-C, cytochrome-C; DNMTs, DNA methyltransferases; GPx, glutathione peroxidase; GPx4, glutathione peroxidase-4; 8-OHdG, 8-hydroxy-2'-deoxyguanosine; H₂O₂, hydrogen peroxide; HO-1, heme oxygenase-1; IL-1β, interleukin-1 beta; IL-6, interleukin-6; iNOS, inducible nitric oxide synthase; LC3a, microtubule associated protein light chain 3A; LC3b-I/II, microtubule associated protein light chain 3B; MDA, malondialdehyde; NQO1, NAD(P)H quinone oxidoreductase 1; Nrf2, nuclear factor erythroid 2-related factor 2; ROS, reactive oxygen species; SOD, superoxide dismutase; TNF-α, tumour necrosis factor-α.

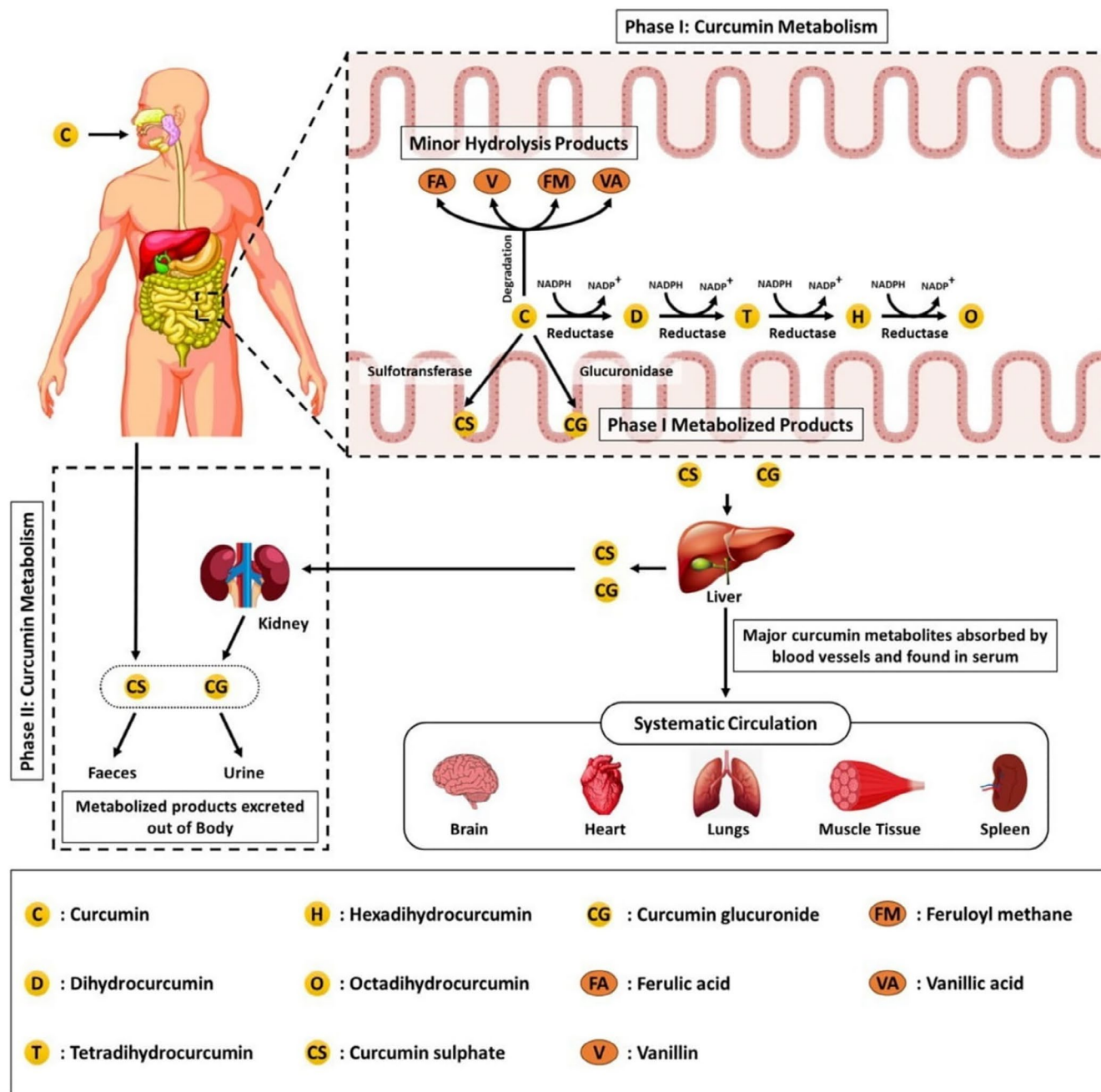


FIGURE 4 | Stages of curcumin metabolism in the human body.

Steward 2005; López-Lázaro 2008). Clinical pilot studies have further linked curcumin intake with regression of premalignant lesions in the stomach, bladder, cervix, skin, and soft palate (Cheng et al. 2001). Additionally, curcumin is reported to exhibit nephroprotective, hepatoprotective, cardioprotective, thrombosis-suppressing, hypoglycemic, anti-inflammatory, anti-Alzheimer, and antirheumatic effects (Anand et al. 2007; Zhou, Beevers, and Huang 2011). Despite these benefits, data on curcumin levels in target tissues remain scarce. It is hypothesised that the observed systemic benefits of curcumin are mediated by its degradation products, metabolites, or natural congeners rather than free curcumin itself. Most pharmacological studies have used turmeric-based preparations containing bisdemethoxy-curcumin and demethoxy-curcumin, which, along with tetrahydro-curcumin (a

reductive metabolite), have been evaluated for their biological activities (Anand et al. 2008). Depending on study endpoints, these derivatives have shown activities comparable to or lower than curcumin. However, the biological potential of other key reductive metabolites, such as octahydrocurcumin and hexahydrocurcumin, remains largely unexplored (Metzler et al. 2013). Comprehensive research is urgently needed to elucidate the roles of curcumin's metabolites and congeners in mediating its health effects.

Degradation products such as vanillin, feruloyl methane, trans-6-(4'-hydroxy-3'-methoxyphenyl)-2,4-dioxo-5-hexenal, and ferulic acid have been suggested to contribute to the pharmacological effects of curcumin (Shen and Ji 2009; Shen and Ji 2012). However, Gordon and Schneider (2012) challenged this

hypothesis, asserting that bicyclopentadione, rather than hexenal, which forms in only minor amounts, is the primary degradation product. Bicyclopentadione, a by-product of curcumin's autoxidative degradation and a metabolite of lipoxygenases and cyclooxygenases, is anticipated to interact significantly with curcumin's biological activity. Examining its pharmacological properties and pharmacokinetic profile is crucial to understanding its potential role in mediating curcumin's health benefits. Given curcumin's poor oral bioavailability, numerous strategies have been explored to enhance its bioavailability. These include developing formulations with adjuvants, liposomes, micelles, polymeric nanoparticles, magnetic nanoparticles, polymeric conjugates, and solid-liquid nanoparticles (El-Saadony et al. 2023).

7 | Safety Aspects of Curcumin

A single heavy dose of curcumin at 5000 mg/kg of body weight was administered orally to the Swiss albino mice, and no toxic effects were observed for 14 days. Similarly, no toxicity was noticed during the 14-day observation period in rats with the same single dose (5000 mg/kg), along with the appearance of no pathological effects and mortality in both mice and rats (Aggarwal, Chacko, and Kuruvilla 2016). Two generations of female and male Wistar rats were treated with curcumin orally at concentrations of 1500 ppm, 3000 ppm, and 10,000 ppm (for the F1 and F0 generations, 10,000 ppm concentration was equivalent to 959 and 847 mg/kg/bd/d for the male rats and 1076 and 1043 mg/kg the body weight/day for the female rats) through dietary intake did not exhibit reproductive toxicity. However, with the highest dose (10,000 ppm), the body weight gain in F2 generation chicks was slightly reduced (Ganiger et al. 2007). The male nude mice were administered with pegylated curcumin intravenously daily for 10 days. The study concluded that testicle weights had no significant change, but the weights of seminal vesicles and levels of testicular testosterone were diminished, and disturbed spermatogenesis was observed (Murphy et al. 2012).

In another study, curcumin was administered orally at 1 g/day for 3 months, and the findings revealed no toxicity in osteoarthritis patients (Pinsornsak and Niempoo 2012). A separate study was conducted on these patients, and 180 mg of curcumin was administered daily through an oral route for 8 weeks. The results showed no major toxicity, but tachycardia and hypertension were reported in a single subject, and another subject showed redness of the tongue (Nakagawa et al. 2014). Several clinical research projects have also reported no major toxicity in osteoarthritis patients treated with a daily dose of 1500 mg curcuminoids taken orally for 6 weeks and in patients receiving 1.5 g/day curcumin for 6 weeks (Panahi et al. 2014; Rahimnia et al. 2015). Intravenous administration of a single dose of liposomal curcumin to 50 healthy female and male subjects was found safe at doses below 120 mg/m²; however, at higher doses, morphological changes in red blood cells were observed (Storka et al. 2015). Some adverse effects, including constipation, headache, and stomach-ache were observed in patients suffering from pulmonary problems caused by sulphur mustard and receiving 1.5 g/day curcumin for 4 weeks, without the appearance of severe adverse effects and major toxicity (Panahi et al. 2015).

8 | Curcumin Administration: Limitations and Future Prospects

Curcumin has a long-established safety record. Both the European Food Safety Authority (EFSA) and The Joint FAO/WHO Expert Committee on Food Additives (JECFA) have set the Allowable Daily Intake (ADI) for curcumin at 0–3 mg/kg of body weight (Kocaadam and Şanlıer 2017). Numerous clinical trials conducted on healthy individuals have consistently supported the safety and efficacy of curcumin. However, despite its generally recognised safety, some adverse effects have been reported. In a dose–response study involving seven subjects who received doses ranging from 500 to 12,000 mg of curcumin, side effects such as headache, rash, yellow stool, and diarrhoea were observed within 72 h (Lao et al. 2006). Similarly, another study involving doses of 0.45–3.6 g/day for 1–4 months reported symptoms of diarrhoea and nausea, along with increases in lactate dehydrogenase and serum alkaline phosphatase levels (Sharma et al. 2004).

Looking ahead, the future of curcumin in improving human health lies in advancing its precision through personalized medicine, exploring resistance mechanisms, investigating its potential therapy synergy with other therapies, and conducting rigorous clinical trials to pave the way for evidence-based integration of curcumin into medical practice (Islam et al. 2024).

9 | Conclusion

Owing to the adverse effects of allopathic medicines, both patients and healthy individuals have transitioned towards herbal remedies. The two primary reasons for this transition are their cost-effectiveness and lesser side effects. Due to extensive industrialization worldwide, the entire ecosystem is loaded with a variety of natural and chemically toxic substances. Prolonged exposure leads to major toxicity in several organs of both humans and animals, including both terrestrial and aquatic. Multiple scientific studies using animal models have demonstrated that curcumin is the best herbal remedy that safeguards against kidney toxicity, liver toxicity, lung toxicity, testicular toxicity and brain toxicity by increasing antioxidant enzyme levels, enhancing defensive signaling pathways, and inhibiting the activities of pro-inflammatory markers etc.

Author Contributions

Harsh Kumar: writing – original draft, writing – review and editing. **Rajni Dhalaria:** writing – review and editing. **Neetika Kimta:** writing – review and editing. **Shivani Guleria:** conceptualization, writing – review and editing. **Navneet Kumar Upadhyay:** conceptualization, writing – review and editing. **Eugenie Nepovimova:** writing – review and editing. **Daljeet Singh Dhanjal:** writing – original draft. **Nidhi Sethi:** writing – review and editing. **Sivakumar Manickam:** conceptualization, supervision.

Consent

All authors gave their consent for publication.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data are available in the articles cited in the review.

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